MODUL UNIVERSITY VIENNA

SEID 2364

SESSION 8

Bias, Mediation, and System Pathways

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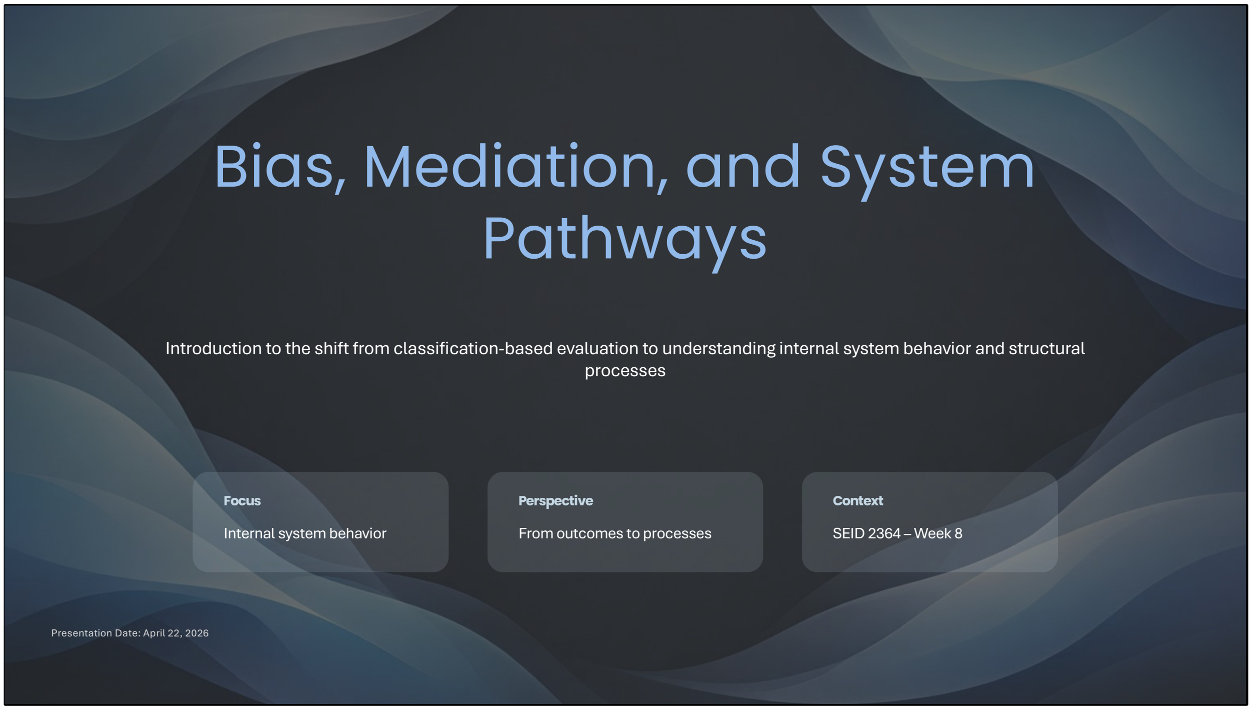
Instructor Notes: Frame this as a structural shift in the course:
• Prior sessions focused on how individuals reason ethically (interpretation, judgment, perspective)
• This session examines what changes when reasoning must be made consistent, repeatable, and transferable across individuals

Emphasize that this is not replacing ethical reasoning, but placing it under constraint.

Intellectual throughline:
• Ethical reasoning produces varied but defensible answers
• Real-world systems require consistent and enforceable decisions
• This tension motivates the need for frameworks

This is the beginning of a move from open-ended reasoning → structured systems of evaluation.



SESSION OVERVIEW: BIAS, MEDIATION, AND SYSTEM PATHWAYS

This slide establishes the session's three analytical lenses and its place within the course progression.

FOCUS: INTERNAL SYSTEM BEHAVIOR
This session marks a fundamental shift in the course's analytical approach. Prior sessions (1–7) examined how individuals reason about ethical questions — interpretation, judgment, perspective-taking. This session turns the lens inward, examining how systems themselves produce outcomes through their internal structures and processes. The focus is not on what systems do (their outputs) but on how they do it (their internal behavior).

PERSPECTIVE: FROM OUTCOMES TO PROCESSES
The shift from outcome-focused to process-focused analysis is the session's core methodological move. Outcome analysis asks "Is this result fair?" Process analysis asks "What structural mechanisms produced this result?" The latter question is more demanding but also more actionable — if you understand the process, you can identify specific intervention points rather than only detecting

problems after they've occurred.

CONTEXT: THE THREE LENSES

The session's three key concepts form a progressive analytical framework:

• Bias — not as a single concept but as a family of phenomena operating across cognitive, statistical, computational, and systemic levels (slides 12–16)

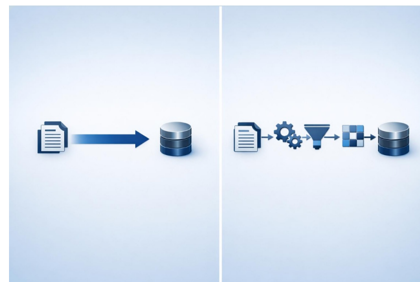
• Mediation — the insight that bias is produced through mediating structures rather than residing in individual components (slides 17–18)

• Pathways — the analytical method of tracing how information transforms as it crosses boundaries between spaces, revealing where bias enters, transforms, and amplifies (slides 19–25)

TEACHING NOTE: Use this slide to set expectations for the session. Students should understand that they will leave with a new analytical tool (the BBS model) and practical experience using it (the bias probe exercise). The session is designed to be both theoretically substantive and practically applicable — every abstract concept will be grounded in concrete examples and hands-on investigation.

From Classification to Mediation

- **Old paradigm:** Classification and evaluation
- **New paradigm:** Mediation and structure
- **Insight:** Processes shape outcomes



FROM CLASSIFICATION TO MEDIATION

This slide presents the paradigm shift that organizes the entire session. Understanding this shift is essential for everything that follows.

OLD PARADIGM: CLASSIFICATION AND EVALUATION
The traditional approach to analyzing AI systems focused on classifying inputs and evaluating outputs:
• Is the data biased or unbiased?
• Is the model accurate or inaccurate?
• Is the outcome fair or unfair?
These binary classifications are useful but limited. They treat system components as isolated objects that can be evaluated independently. They don't explain how bias moves through a system, changes form, or compounds across stages.

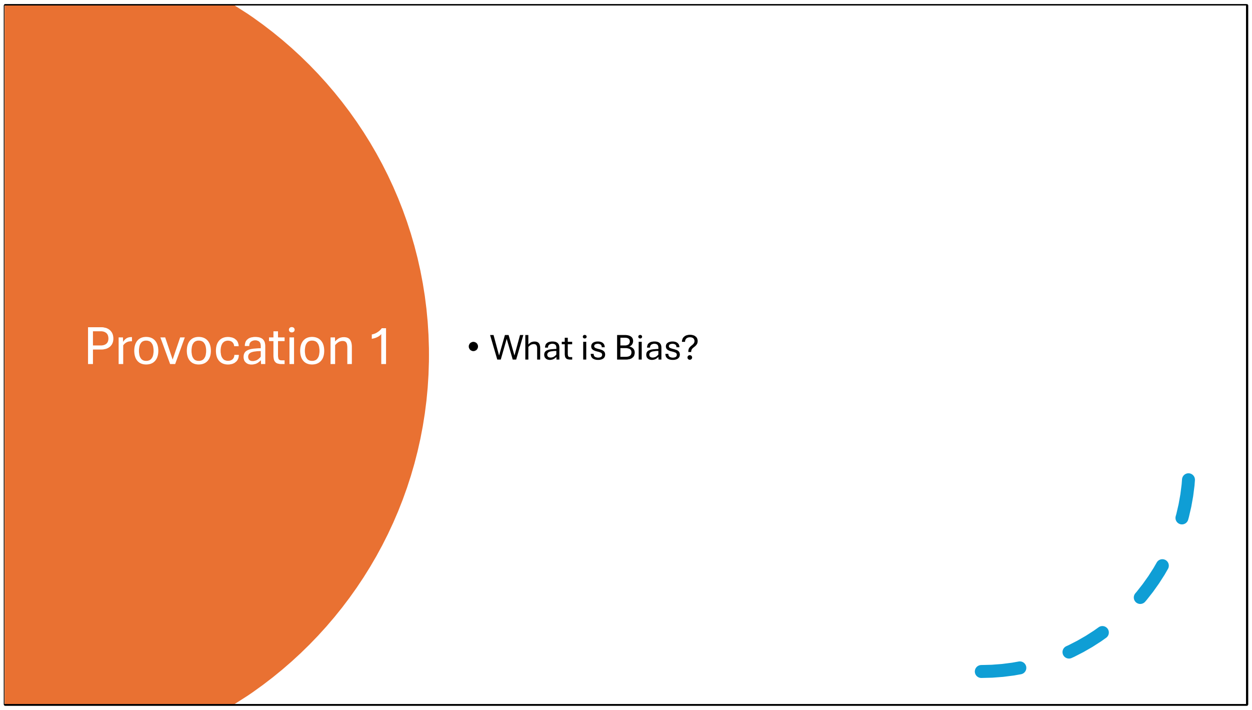
NEW PARADIGM: MEDIATION AND STRUCTURE
The mediation paradigm shifts the analysis from components to pathways:
• Instead of asking "is the data biased?" ask "how does information transform as

it crosses the data collection boundary, and who is advantaged or disadvantaged by that transformation?"
• Instead of asking "is the model fair?" ask "what structural mechanisms within the model's processing pathway produce differential outcomes for different groups?"
• Instead of evaluating endpoints, trace the full journey of information through the system

INSIGHT: PROCESSES SHAPE OUTCOMES
This is the foundational claim: you cannot understand outcomes without understanding the processes that produce them. And processes are not neutral pipes — they are active transformers that shape, filter, and redirect information at every stage. Each transformation point (each boundary crossing in BBS terms) is both a functional necessity (the system needs to process information) and a potential bias site (the transformation may systematically advantage some and disadvantage others).

Outcomes come from processes — understanding system behavior requires examining the mediating structures that transform inputs into outputs. Each location within any mediation process is a potential point for ethical evaluation.

TEACHING NOTE: This slide is a conceptual anchor — return to it throughout the session. When discussing specific bias types (slides 12–16), point out how each type relates to mediation. When discussing the BBS model (slides 18–19), show how it operationalizes the mediation paradigm. When conducting the exercise (slides 26–29), remind students that they are applying mediation analysis, not just testing inputs and outputs.



PROVOCATION 1: WHAT IS BIAS?

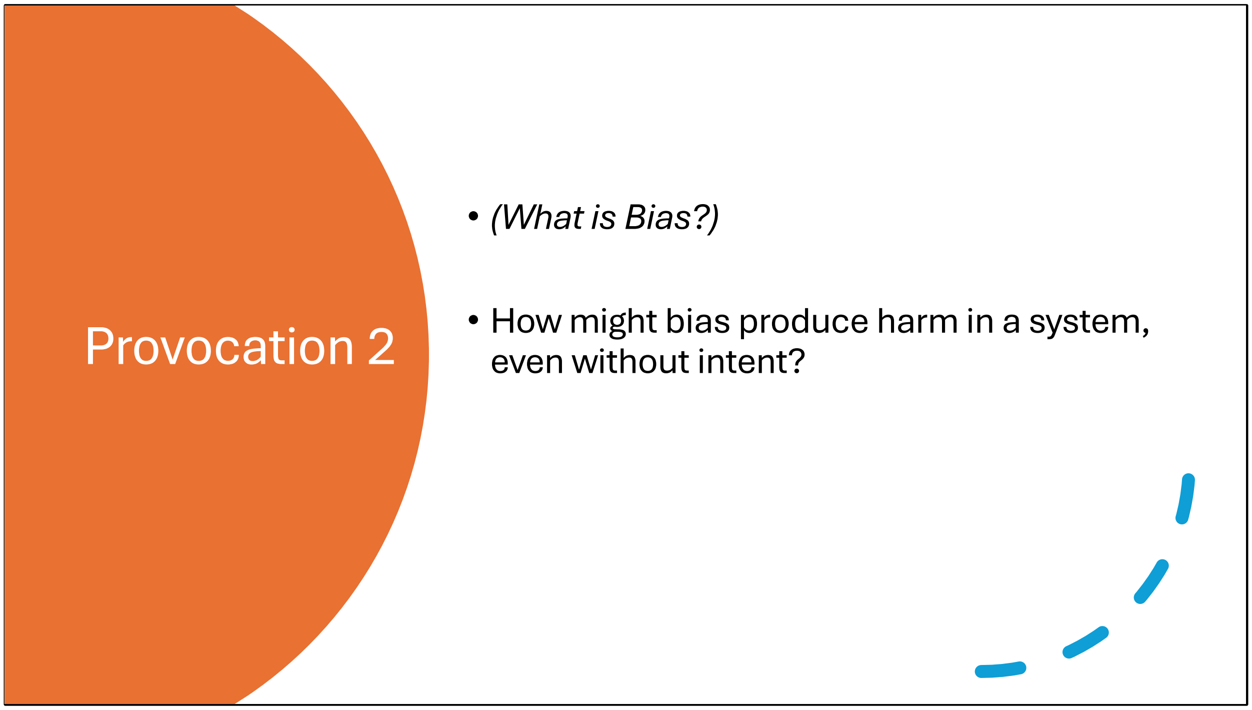
Purpose: This provocation is designed to surface students' pre-existing mental models of bias before the session provides formal frameworks. The goal is not to reach a consensus definition but to reveal the diversity of ways the concept is understood — and to use that diversity as evidence for why careful conceptual work is necessary.

Facilitation guidance:
• Give students 60–90 seconds of silent thinking time before opening discussion. This prevents the first confident voice from anchoring the room.
• Expect answers ranging from everyday language ("being unfair," "having a preference") to more technical framings ("systematic error," "skewed data"). Both are valid starting points.
• Track responses on the board in rough clusters — you'll return to these clusters when slides 12–16 formally distinguish human, statistical, early systems, ML, and contemporary AI bias.
• Push for specificity: when a student says "bias means being unfair," ask "unfair to whom? measured how? compared to what baseline?" This plants the seed for

the statistical and structural framings that come later.

Common student misconceptions to watch for:
• "Bias is always intentional" — this session will systematically dismantle this assumption
• "Bias is a property of data" — the session will show bias as a property of pathways, not components
• "Removing bias means being neutral" — the BBS model will show that every encoding choice is a bias-generating decision, and neutrality is itself a position

Intellectual setup: This provocation connects to the session's arc by establishing that "bias" is not a single concept but a family of related phenomena that operate differently across cognitive, statistical, computational, and systemic contexts. The progression through slides 12–16 will formalize what students intuitively gesture at here.



PROVOCATION 2: HOW CAN BIAS PRODUCE HARM IN A SYSTEM, EVEN WITHOUT INTENT?

Purpose: This provocation shifts students from thinking about bias as a personal moral failing to thinking about it as a structural property of systems. The phrase "even without intent" is the key pedagogical move — it forces students to consider mechanisms of harm that don't require a villain.

Facilitation guidance:
• This is often a more challenging provocation than the first because it requires students to think systemically rather than individually. Give them more time.
• If students struggle, offer a concrete anchor: "Imagine a hiring algorithm trained on ten years of hiring data from a company that historically hired mostly men. No one programmed it to prefer men. Does it produce harm? How?"
• Push students to identify the mechanism, not just the outcome. "The algorithm discriminates" is an observation. "The algorithm learned patterns from historically biased data and reproduces those patterns at scale" is an analysis.
• Track whether students can distinguish between: (a) bias that produces harm because someone intended it, (b) bias that produces harm because no one

noticed it, and (c) bias that produces harm because it's a structural property of the system's design.

Key concepts this provocation sets up:
• Structural harm (slide 9) — harm that operates through system architecture rather than individual action
• Distributed emergence (slide 11) — bias as a system-level property rather than a component-level defect
• The distinction between authored and emergent bias (slides 14–15) — the progression from explicit rules to learned patterns

Connection to course arc: This provocation operationalizes the move from individual ethical reasoning (sessions 1–7) to systemic analysis. When bias can produce harm without intent, individual moral responsibility becomes insufficient as an analytical framework — you need structural analysis tools like the BBS model.

Provocation 3

- *(What is Bias?)*
- *(How might bias produce harm in a system, even without intent?)*
- Using your Assignment 7 system
- Describe where bias can enter, transform, or amplify in ways that could produce uneven outcomes

PROVOCATION 3: USING YOUR ASSIGNMENT 7 SYSTEM — DESCRIBE WHERE BIAS CAN ENTER, TRANSFORM, OR AMPLIFY

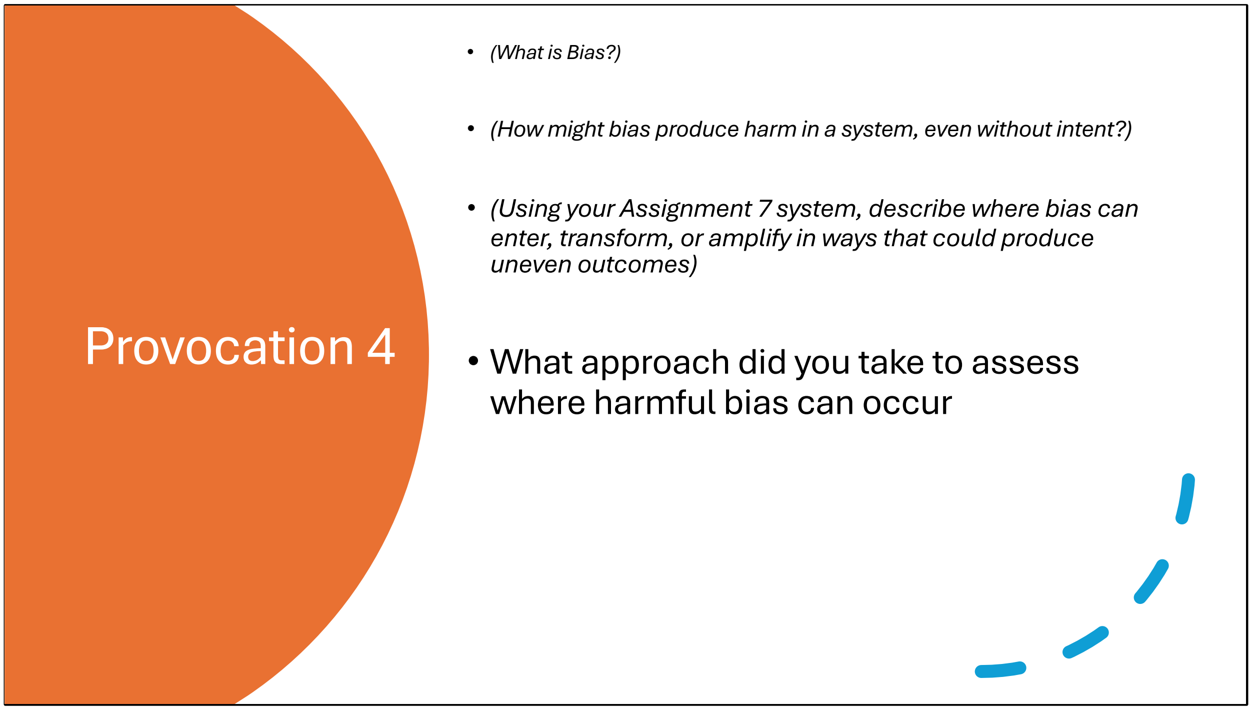
Purpose: This provocation grounds the abstract concept of systemic bias in students' own work. By asking them to analyze their Assignment 7 systems specifically, it forces the conceptual move from "bias is a thing that exists somewhere" to "bias enters, transforms, and amplifies through identifiable pathways in a specific system."

Facilitation guidance:
• The three verbs — enter, transform, amplify — are deliberately chosen and sequential. Coach students to address each separately:
  - ENTER: Where does bias first get into the system? Through training data? User inputs? Design assumptions? API choices?
  - TRANSFORM: How does bias change form as it moves through the system? A demographic proxy in input data becomes a weighted feature becomes a threshold decision — the bias is "the same" but its form has shifted at each stage.
  - AMPLIFY: Where do feedback loops, compounding effects, or threshold

decisions take a small initial bias and make it larger? This is where the most dangerous systemic effects live.
• Students often identify entry points easily but struggle with transformation and amplification. Push them on these.
• Have students sketch a simple pathway diagram of their system — even rough boxes and arrows help them think structurally.

Connection to later content:
• This provocation directly sets up slide 10 (the expanded discussion of entry, transformation, and amplification)
• It also prepares students for the exercise on slides 26–29 where they'll systematically probe their systems
• The BBS model (slide 18) will give them formal vocabulary for the pathway analysis they're doing intuitively here

TEACHING NOTE: If students say "I don't think my system has bias," redirect rather than correct. Ask: "What assumptions does your system make about its users? What information does it discard? What categories does it impose?" Every system has these, and each is a potential bias pathway.



PROVOCATION 4: WHAT APPROACH DID YOU TAKE TO ASSESS WHERE HARMFUL BIAS CAN OCCUR?

Purpose: This is a meta-cognitive provocation — it asks students not just what they found but how they looked. The shift from "what is bias?" to "how do you find bias?" is the shift from conceptual understanding to analytical methodology. This provocation surfaces students' implicit analytical frameworks so they can be examined, compared, and improved.

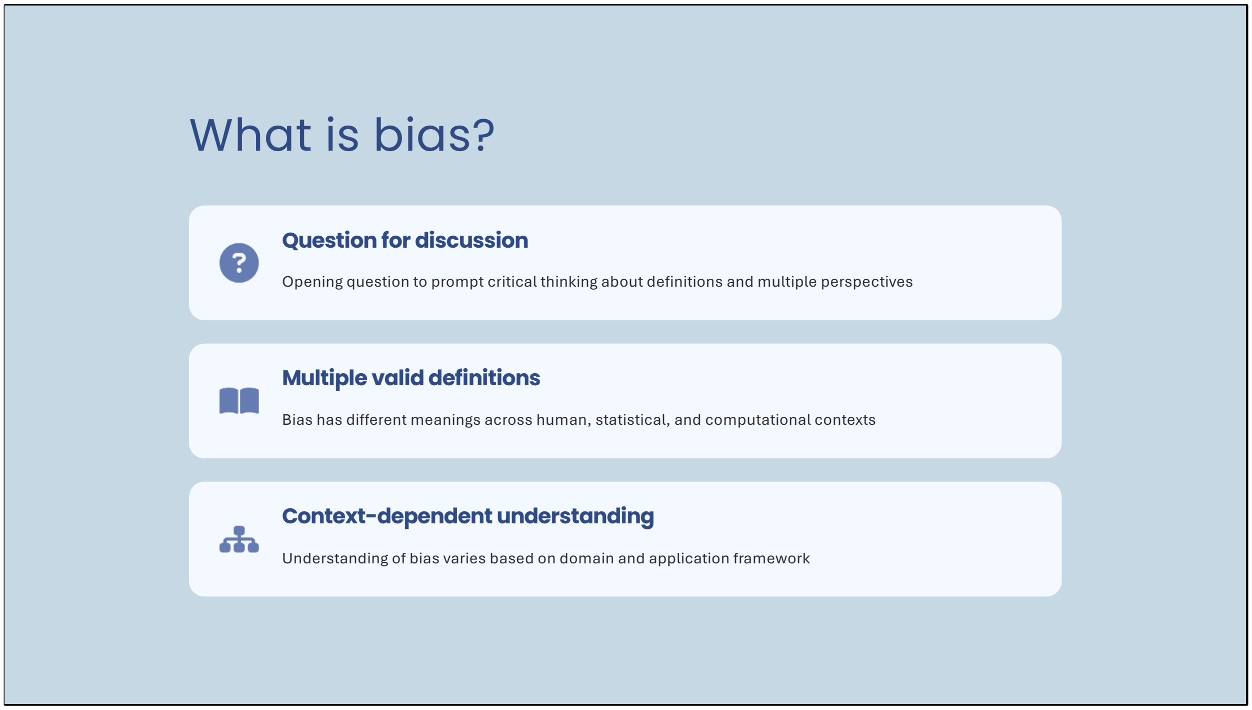
Facilitation guidance:
• Expect a wide range of approaches: some students will have tested inputs systematically, others will have read documentation, others will have reasoned from first principles about data flows.
• Categorize approaches on the board as they emerge. Common categories include:
  - Input testing: "I tried different names/demographics and compared outputs"
  - Documentation review: "I read what data the system uses and looked for known biases"
  - Structural analysis: "I traced how information flows through the system and

identified transformation points"
  - Outcome comparison: "I looked at whether the system produces different results for different groups"
• Each approach has strengths and blind spots. Input testing finds behavioral bias but misses structural causes. Documentation review identifies known issues but not emergent ones. The goal is not to crown one approach as correct but to show that comprehensive bias assessment requires multiple methods.

Key pedagogical move: After students share their approaches, point out that most focused on either inputs or outputs — very few examined the transformation pathway in between. This is precisely the gap that the BBS model fills. The model provides a systematic framework for examining what happens between input and output, which is where most bias actually operates.

Connection to course arc:
• This provocation sets up the exercise methodology on slides 26–29
• It motivates the BBS framework by showing students the limitations of their intuitive approaches
• It connects to slide 32's question about consistent evaluation — if everyone uses different methods to find bias, how do we evaluate systems consistently?

Discussion



WHAT IS BIAS? — EXPANDED DISCUSSION SLIDE

Purpose: This slide transitions from the open provocation (slide 4) to structured analysis. It establishes three foundational points: bias has multiple valid definitions, those definitions are context-dependent, and the context of this course requires holding all of them simultaneously.

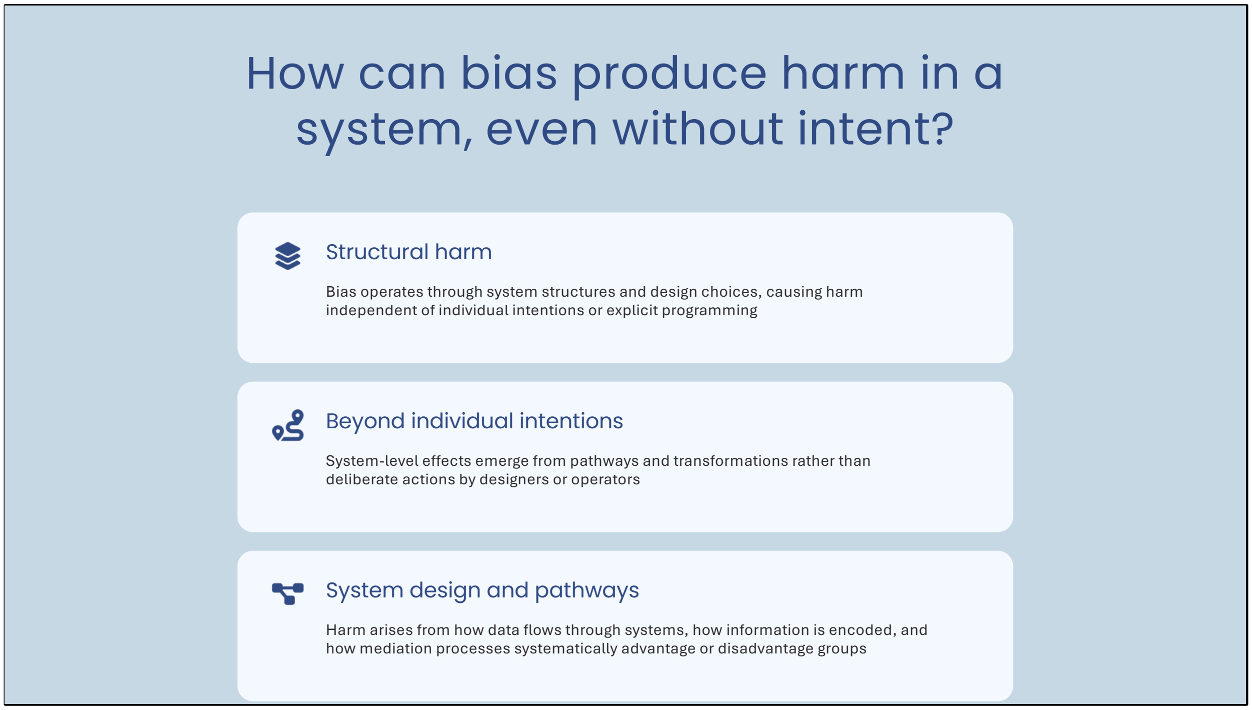
QUESTION FOR DISCUSSION
This is not a rhetorical question — it's designed to surface the definitional plurality that students encountered in the provocation. By now, the board should have multiple clustered responses from Provocation 1. Use this slide to organize those clusters into the three frameworks the next slides will explore.

MULTIPLE VALID DEFINITIONS
This is a critical conceptual point. Students often assume that "bias" has one correct definition and the task is to find it. Instead, emphasize that:
• In cognitive science, bias refers to systematic patterns in human perception and judgment (expanded on slide 12)
• In statistics, bias refers to systematic deviation from true values (expanded on

slide 13)
• In early computing, bias was explicitly encoded in rules and thresholds (expanded on slide 14)
• In modern ML, bias is an emergent property of learned patterns (expanded on slide 15)
• In contemporary AI systems, bias is a distributed system behavior (expanded on slide 16)
None of these definitions is wrong. They operate at different levels of analysis and apply to different system components. The challenge — and the point of this session — is that real-world AI systems contain all of these simultaneously.

CONTEXT-DEPENDENT UNDERSTANDING
The meaning of "bias" shifts depending on your analytical framework, your disciplinary background, and the system component you're examining. A data scientist, a cognitive psychologist, a civil rights attorney, and a system designer would all define bias differently — and all would be correct within their domain. The BBS model (introduced later) provides a way to integrate these perspectives by locating each type of bias at specific points in the system pathway.

TEACHING NOTE: This slide is a bridge, not a destination. Don't spend more than 3–4 minutes here. Its purpose is to validate the complexity students discovered in the provocation and set up the structured progression through slides 12–16 where each type of bias gets full treatment.



HOW CAN BIAS PRODUCE HARM IN A SYSTEM, EVEN WITHOUT INTENT? — EXPANDED DISCUSSION

Purpose: This slide unpacks the core insight from Provocation 2 into three analytical dimensions: structural harm, intentions vs. system effects, and the role of system design and pathways.

STRUCTURAL HARM

On the slide: "Bias operates through system structures and design choices, causing harm independent of individual intentions or explicit programming"
Expanded explanation: Structural harm is harm that is produced by the way a system is organized rather than by the deliberate actions of any individual within it. The classic example is redlining: no single bank teller decided to discriminate, but the system of mapped neighborhoods, risk categories, and lending thresholds produced systematic exclusion of Black communities. The harm was real, measurable, and devastating — but it didn't require any individual actor to harbor racist intent. It was built into the structure.
In AI systems, structural harm works the same way. A facial recognition system that performs poorly on darker skin tones doesn't do so because a programmer

wrote racist code. It does so because the training data over-represented lighter skin tones, the evaluation metrics didn't disaggregate by skin tone, and the deployment context didn't include performance audits across demographics. Each of these is a structural choice, not an intentional one.

BEYOND INDIVIDUAL INTENTIONS

On the slide: "System-level effects emerge from pathways and transformations rather than deliberate actions by designers or operators"

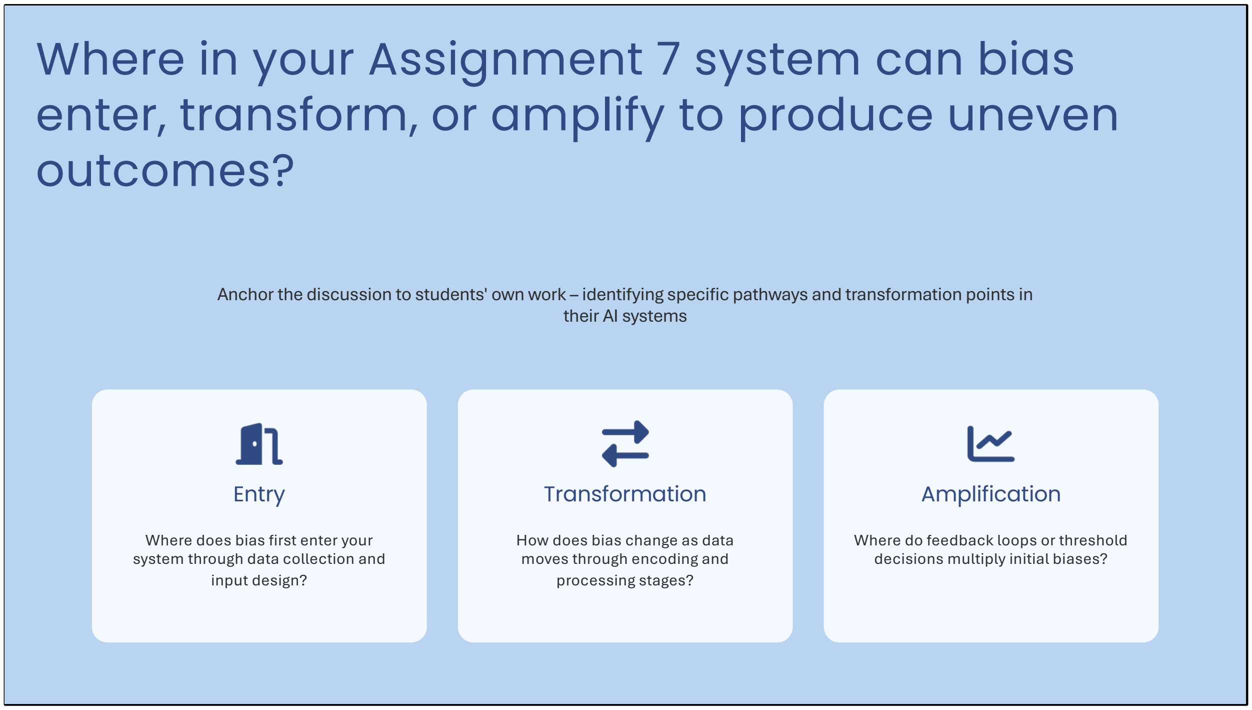
Expanded explanation: This is the key conceptual shift students need to make. Our default moral reasoning framework assigns responsibility based on intent: if you meant to cause harm, you're culpable. But systemic bias breaks this framework because the harm emerges from the interaction of many well-intentioned decisions. The data scientist who chose a standard dataset, the engineer who optimized for aggregate accuracy, the product manager who prioritized speed to market — none intended harm, but their collective decisions produced it. This is why pathway analysis (BBS model) is necessary: you need to trace how individually reasonable choices compound into systematically harmful outcomes.

SYSTEM DESIGN AND PATHWAYS

On the slide: "Harm arises from how data flows through systems, how information is encoded, and how mediation processes systematically advantage or disadvantage groups"

Expanded explanation: This introduces the mechanism that the rest of the session will formalize. Harm is not a property of any single component but of the pathway — the sequence of encodings, transformations, and decisions that information passes through. At each step, certain representations are privileged and others are suppressed. Certain groups' experiences align with the system's categories and others don't. The cumulative effect of these small, structural advantages and disadvantages is systematic inequality in outcomes.

TEACHING NOTE: Use a concrete example to anchor this slide. The loan system example (slides 21–25) works well here as a preview: "We'll look at this in detail later, but imagine a loan application system. No one programmed it to discriminate by race. But the form design, the data encoding, the model training, and the threshold setting each introduce small structural advantages for certain groups. By the time the system produces a decision, those small advantages have compounded into measurable disparities. That's structural harm through pathways."



WHERE IN YOUR ASSIGNMENT 7 SYSTEM CAN BIAS ENTER, TRANSFORM, OR AMPLIFY TO PRODUCE UNEVEN OUTCOMES?

Purpose: This slide provides the structured framework for Provocation 3's discussion. It decomposes the bias pathway into three distinct phases — entry, transformation, amplification — and asks students to locate each in their own systems.

ENTRY: WHERE DOES BIAS FIRST ENTER YOUR SYSTEM?
Expanded explanation: Bias entry points are the locations where real-world complexity first encounters the system's representational constraints. Common entry points include:
• Data collection: What gets measured and what doesn't? Whose experiences are represented in training data? What populations were excluded from data collection?
• Input design: What fields are on the form? What dropdown options are available? What categories are imposed? What information is marked "required" vs. "optional"?
• Labeling and annotation: Who labeled the training data? What instructions did

they receive? What implicit assumptions guided their categorization?
• API and system boundaries: What data does the system accept from external sources? What format constraints does it impose? What gets rejected or truncated?
Each entry point is a boundary crossing in BBS terms — a place where the phenomenological reality of a person or situation gets translated into a system-readable representation.

TRANSFORMATION: HOW DOES BIAS CHANGE AS DATA MOVES THROUGH THE SYSTEM?
Expanded explanation: Once bias enters a system, it doesn't stay in its original form. It transforms as data moves through processing stages:
• Encoding transforms qualitative information into quantitative representations (a person's work history becomes a numerical feature)
• Normalization changes the scale and distribution of data, potentially obscuring or amplifying group differences
• Feature engineering combines and recombines variables, creating proxy variables that carry bias even when protected attributes are removed
• Model training optimizes for aggregate performance, potentially sacrificing accuracy for minority subgroups
The transformation phase is where bias becomes invisible — it's no longer recognizable as the original design choice that introduced it. This is why component-level auditing (checking the data, checking the model) often misses systemic bias: the bias has transformed into a different form at each stage.

AMPLIFICATION: WHERE DO FEEDBACK LOOPS OR THRESHOLD DECISIONS MULTIPLY INITIAL BIASES?
Expanded explanation: Amplification is where small initial biases become large systemic effects. Key amplification mechanisms include:
• Feedback loops: System outputs influence future inputs (predictive policing → more arrests in targeted areas → more data confirming the prediction)
• Threshold effects: Binary decisions (approve/deny, flag/pass) convert continuous bias into discrete exclusion
• Compounding: Multiple biased components in sequence multiply their effects (biased data + biased model + biased threshold = dramatically biased outcome)
• Scale: Automated systems apply bias consistently across millions of decisions, turning what might be occasional human inconsistency into systematic discrimination

TEACHING NOTE: Have students map their Assignment 7 system onto these three categories. The most valuable learning moment is often when students realize that a bias they identified as an "entry" problem is actually an "amplification" problem — or that what they thought was a single bias is actually three different biases operating at different pathway stages. Push for this level of specificity.

Bias as System Behavior

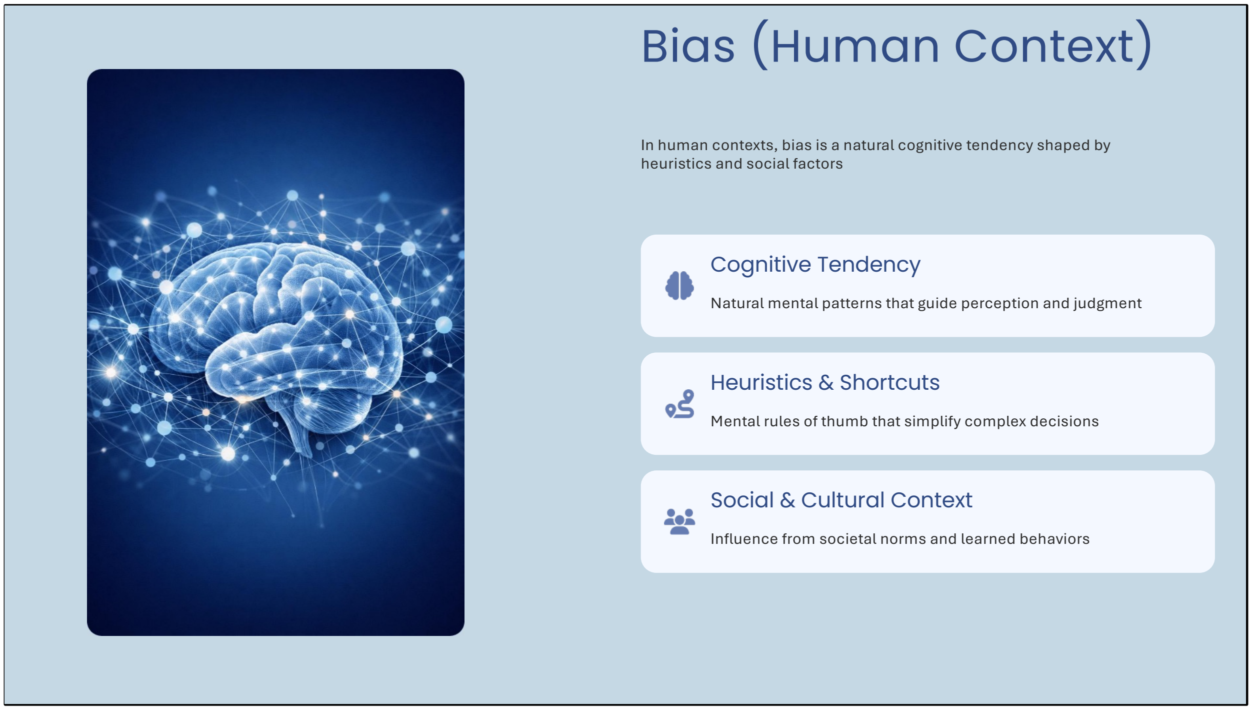
Bias is not isolated – it emerges through structure and distributed processes across the entire system

Not: Single point failure

Instead: Distributed emergence

Mechanism: Structural pathways

"Not: Single point failure" — Bias doesn't originate from one broken component (a bad dataset, a flawed algorithm, or a biased designer acting alone).
"Instead: Distributed emergence" — Bias arises from the *interaction* of many components across the system — data collection, encoding, model architecture, feedback loops, deployment context.
"Mechanism: Structural pathways" — The BBS model (introduced earlier in the deck) shows how information transforms as it crosses boundaries between spaces, and bias can enter or amplify at *each* of those transitions.

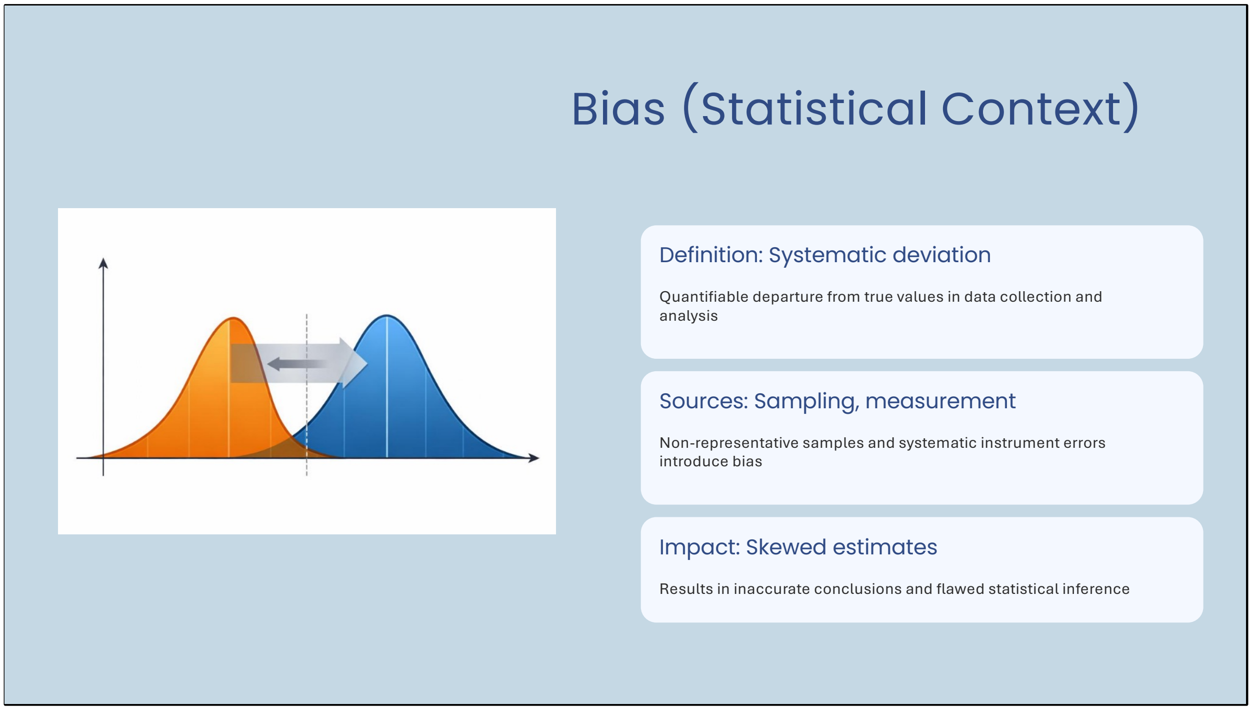


Cognitive Tendency *On the slide:* "Natural mental patterns that guide perception and judgment"
Clearer explanation: Everyone's brain takes shortcuts when processing information. Cognitive biases are systematic patterns in how we perceive, remember, and interpret things — they're not random errors but predictable tendencies built into how human cognition works. For example, confirmation bias leads us to favor information that supports what we already believe, while anchoring bias causes us to over-rely on the first piece of information we encounter. These aren't flaws per se — they evolved to help us make fast decisions — but they can distort judgment, especially in complex or unfamiliar situations.

Heuristics & Shortcuts *On the slide:* "Mental rules of thumb that simplify complex decisions"
Clearer explanation: Heuristics are the brain's efficiency strategies — quick, practical rules we apply when a full analysis isn't feasible (which is most of the time). Think of them as mental "good enough" algorithms. The availability heuristic, for instance, makes us judge something as more likely if we can easily recall examples of it. The representativeness heuristic makes us categorize

things based on how closely they match a prototype. These shortcuts work well in everyday life, but they become dangerous in system design — when a designer's heuristic assumptions get baked into rules, thresholds, or categories, they silently encode bias into the system's structure.

Social & Cultural Context *On the slide:* "Influence from societal norms and learned behaviors"
Clearer explanation: Beyond individual cognition, bias is also shaped by the social and cultural environments we grow up in and operate within. This includes implicit associations (e.g., unconsciously linking certain demographics with certain traits), institutional norms that define what's "normal" or "standard," and cultural narratives that frame how we understand groups of people. This layer is critical for AI ethics because when humans design systems, collect data, or label training sets, these socially learned biases come along — often invisibly. The bias isn't just "in the data" — it's in the worldview of the people who structured the data collection in the first place.



DEFINITION: SYSTEMATIC DEVIATION
Statistical bias is a systematic — not random — departure from the true value. The key word is systematic: random error averages out over time, but bias doesn't. It pulls every estimate in the same direction, consistently. This means more data doesn't fix it — a biased instrument or sampling method will just produce a more precise wrong answer. This is fundamentally different from noise. When students hear "bias," they often think "error," but statistical bias is specifically directional error that persists across repeated measurements.

SOURCES: SAMPLING, MEASUREMENT — AND BEYOND
The slide names the two classic sources, but there are deeper structural ones worth discussing:

• Sampling bias — your sample doesn't represent the population. In AI contexts this is often invisible because the "population" itself is poorly defined.

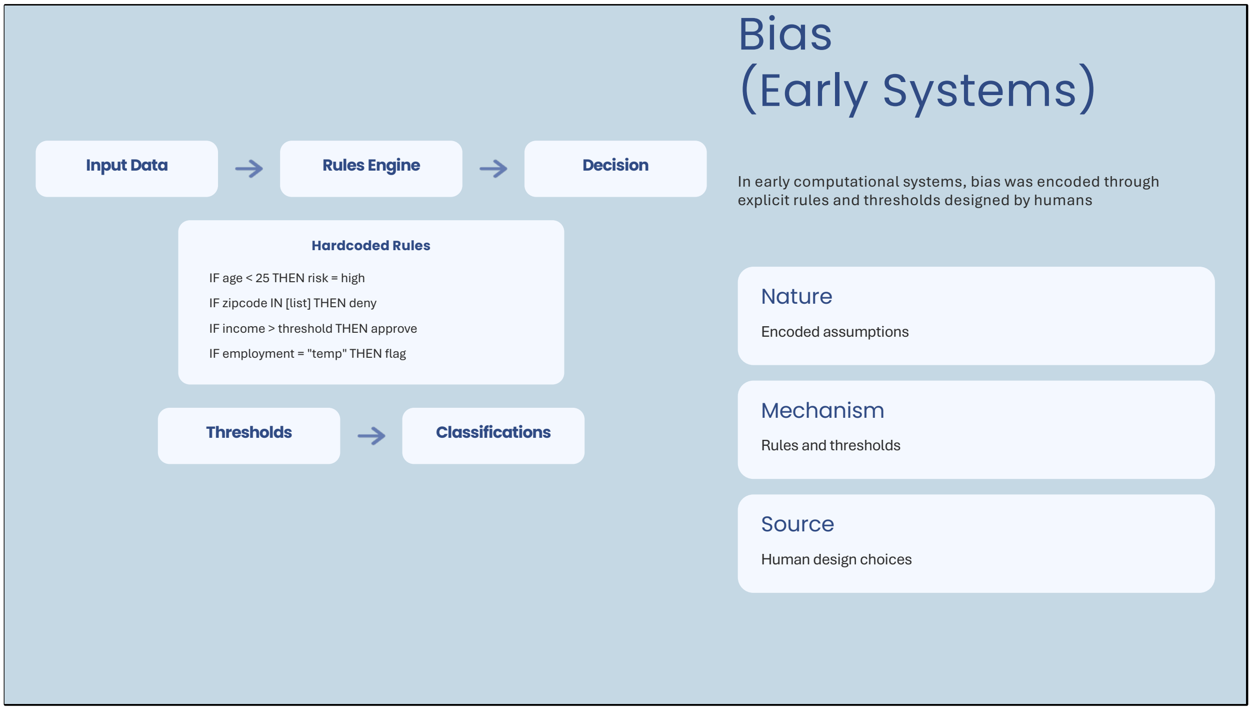
• Measurement bias — your instruments systematically distort what they capture. A sensor that drifts, a survey question that leads.

• Signal degradation and information loss — Data doesn't just "exist" — it's transmitted, compressed, stored, and re-encoded. At each step, fidelity can degrade. Inadequate error correction during transmission, lossy compression, rounding, or format conversion all introduce systematic distortion that compounds downstream. This isn't classical "bias" in the textbook sense, but it produces the same structural effect: a consistent departure from the true state.

• Conceptual space transformation — This is the most important one for this course. When data crosses from one representational context to another — say, from a clinical observation to a database field, or from a natural language response to a numerical encoding — you're not just losing information. You're changing the conceptual space in which the information exists. A qualitative judgment becomes a number. A spectrum becomes a category. The relationships between data points, and even what counts as "close" or "similar," are redefined by the new space. This is where statistical bias connects directly to the BBS model: every boundary crossing is a potential conceptual space transformation, and each one can introduce systematic distortion that's invisible if you only look within a single space.

IMPACT: SKEWED ESTIMATES — AND STRUCTURAL MISREPRESENTATION
The downstream effect isn't just "wrong numbers." When bias compounds through a pipeline — sampling distortion, plus measurement error, plus signal degradation, plus conceptual space shifts — the final output can be structurally misaligned with reality in ways that no amount of statistical correction within the output space can fix. The estimate isn't just skewed; it's answering a subtly different question than the one you think you're asking. This is why pathway-based analysis matters: you need to trace where the distortion entered and how the conceptual space changed, not just calibrate the endpoint.

TEACHING NOTE:
This slide bridges from the human cognitive framing (slide 12) into the formalization of bias in measurement systems. Emphasize that once bias enters a representational system, it becomes compoundable in ways that individual cognitive bias is not. The conceptual space point foreshadows exactly what happens at the boundary objects in the BBS model — each crossing is where the nature of the information fundamentally changes, not just its precision.



HARDCODED RULES: BIAS BY DESIGN

In early computational systems, bias wasn't emergent or hidden — it was explicitly authored. Designers wrote rules like "IF age < 25 THEN risk = high" or "IF zipcode IN [list] THEN deny" directly into decision engines. These weren't bugs; they were features. Someone sat down, made assumptions about which attributes predict which outcomes, and encoded those assumptions as deterministic logic. The bias here is fully traceable to a human design choice — you can point to a line of code and say "this is where the discrimination lives." That traceability is actually what makes early systems useful as a teaching contrast: the bias is visible in ways it won't be in later paradigms.

THE RULE-THRESHOLD MECHANISM

The core mechanism in these systems is the combination of rules and thresholds. Rules define the logic ("if X, then Y"), and thresholds define the boundaries ("where does X become 'too much' or 'not enough'?"). Both embed assumptions:

• Rules encode which variables matter — and by omission, which don't. Choosing to include zipcode but not wealth, or employment type but not employment history, is itself a bias-generating decision.

• Thresholds encode where lines get drawn — and those lines are rarely neutral.

An income threshold of $50,000 vs. $35,000 includes or excludes entirely different populations. A "temp employment = flag" rule treats an entire category of workers as suspect, regardless of individual context.
• The interaction between rules creates compound effects. No single rule may look discriminatory, but the combination — age AND zipcode AND employment type — can produce systematic exclusion of specific demographic groups without any single rule naming those groups.

ENCODED ASSUMPTIONS: THE INVISIBLE WORLDVIEW

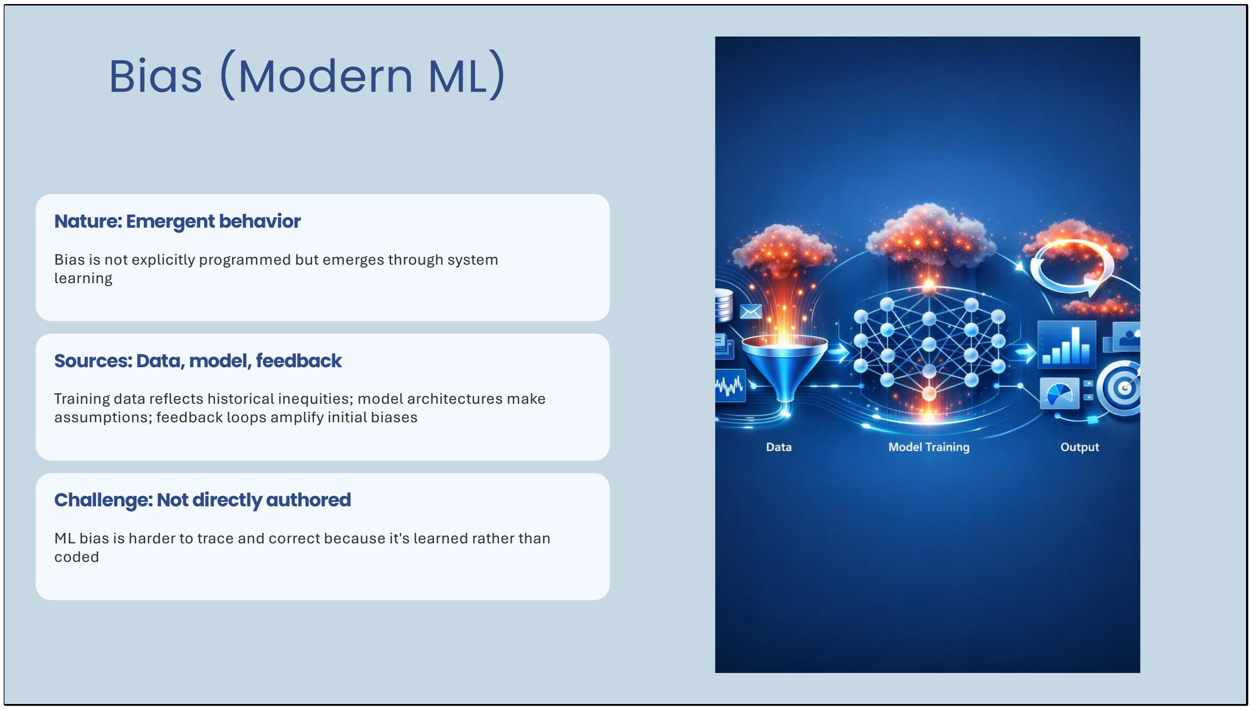
Every rule in these systems reflects a worldview about what's "normal," "risky," or "acceptable." These assumptions often came from historically discriminatory practices — redlining maps became zipcode lists, gendered employment norms became employment-type flags. The system doesn't "know" it's discriminating; it's faithfully executing the logic it was given. This is a key point for students: the system has no intent, but it produces intentional outcomes because the intent was embedded at design time.

CONNECTION TO THE BBS MODEL

In BBS terms, early systems have very simple pathways — the boundary objects are the intake forms, the left virtual space is the rule engine's data structure, and the right virtual space is the binary decision output. But because the transformation is explicit (rules you can read), this is the clearest case for students to practice pathway analysis. Ask them: where does the real applicant (LP) lose dimensionality? At the form (boundary object). Where does bias enter? At the rule definitions (LV). Where does it become consequential? At the threshold-based decision (RV). This slide is essentially the BBS model in its simplest possible form — use it to build fluency before the complexity of ML and contemporary AI in the following slides.

TEACHING NOTE:

This is the pivot point in the bias progression. Slide 12 showed bias as a human cognitive tendency. Slide 13 showed how it formalizes in measurement. This slide shows what happens when formalized bias gets automated — it becomes scalable, consistent, and invisible to the people affected by it. The examples on the slide (age, zipcode, income, employment type) are deliberately chosen to echo real historical practices like redlining and credit discrimination. Push students to see that "the algorithm decided" is never the full story — someone decided what the algorithm would decide.



NATURE: EMERGENT BEHAVIOR

This is the fundamental shift from early systems. In rule-based systems, bias was authored — someone wrote it into the code. In modern ML, bias is *learned*. No one writes "IF female THEN pay less." Instead, the model ingests historical data where women were paid less, discovers that gender (or its proxies) correlates with compensation outcomes, and reproduces that pattern in its predictions. The bias is real and systematic, but it was never explicitly programmed. This is what makes ML bias qualitatively different and far harder to address — you can't grep the codebase for the discriminatory line, because it doesn't exist as a line. It exists as a distributed pattern across millions of weighted parameters.

SOURCES: DATA, MODEL, FEEDBACK

The slide names three sources, and each deserves unpacking:

• Data bias — Training data reflects the world that produced it, including all of that world's historical inequities. A hiring model trained on ten years of hiring decisions will learn the biases of the humans who made those decisions. A medical model trained on clinical data will learn the diagnostic disparities built into who gets tested, who gets referred, and who gets treated. The data isn't "wrong" in a statistical sense — it accurately represents a biased reality. That's

precisely the problem.
• Model bias — The architecture itself makes assumptions. What loss function you optimize, how you handle missing data, what features you engineer, how you regularize — each choice shapes what the model can and cannot learn. A model optimized for overall accuracy will sacrifice performance on minority subgroups if that trade-off improves the aggregate metric. The choice of what to optimize *is* a bias decision, even when it looks purely technical.
• Feedback bias — This is often the most dangerous and least understood source. When a model's outputs influence the data it will later be trained on, you get feedback loops that amplify initial biases. A predictive policing model sends more officers to certain neighborhoods, which produces more arrests in those neighborhoods, which confirms the model's prediction, which sends even more officers. The bias doesn't just persist — it compounds with each cycle. Initial small distortions become structural features of the system.

CHALLENGE: NOT DIRECTLY AUTHORED

The core challenge for governance, accountability, and intervention is that ML bias is not directly authored. You can't point to a decision-maker and say "you wrote this rule." The bias emerged from the interaction of data, architecture, and optimization — a process that no single human fully controlled or even fully understood. This creates what we might call an *accountability gap*: the system produces discriminatory outcomes, but the causal chain is distributed across thousands of design micro-decisions, data collection choices, and mathematical optimizations.

This is also where the conceptual space transformation from slide 13 becomes critical again. When a person's life circumstances get encoded as a feature vector, that encoding is a boundary crossing in the BBS sense. And when the model transforms that feature vector through hidden layers into a prediction, each layer is another conceptual space shift — one that's mathematically precise but semantically opaque. You can compute exactly what the model did, but you often can't explain *why* in human-meaningful terms.

CONNECTION TO THE PROGRESSION

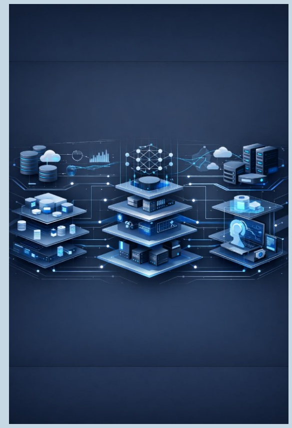
Walk students through the arc so far: slide 12 showed bias as a cognitive tendency (individual, recognizable, potentially correctable through awareness). Slide 13 showed it formalized in measurement (systematic, compoundable, but still traceable). Slide 14 showed it automated through rules (scalable and consistent, but still readable). Now slide 15 shows it learned through data (scalable, consistent, and *opaque*). Each step in this progression makes bias harder to locate, harder to attribute, and harder to correct — which is exactly why the BBS pathway analysis becomes more important, not less, as systems become more complex.

TEACHING NOTE:

Students often want to solve ML bias by "fixing the data" — removing protected attributes, balancing datasets, etc. Push back on this. Removing gender from a dataset doesn't remove gender bias if the model can reconstruct gender from proxies (zipcode, purchasing patterns, name). Balancing a dataset doesn't fix the underlying measurement biases in how the data was collected. The intervention has to happen at the pathway level — understanding *where* and *how* information transforms, not just what goes in and what comes out. This sets up the contemporary AI framing on the next slide, where bias becomes fully distributed across system layers.

Bias (Contemporary AI)

- **Nature:** Distributed system behavior emerging across multiple layers
- **Mechanism:** Operates through mediation layers in complex pathways
- **Framework:** Pathway-based analysis reveals interaction patterns



NATURE: DISTRIBUTED SYSTEM BEHAVIOR EMERGING ACROSS MULTIPLE LAYERS

This is the culmination of the progression. In contemporary AI systems, bias is no longer located in any single component — not in the data, not in the model, not in the rules, not in any one designer's choices. It's a *system-level property* that emerges from the interaction of all components simultaneously. Think of it like weather: you can't point to one molecule and say "that's the storm." The storm is a pattern that emerges from the interaction of millions of elements across multiple atmospheric layers. Contemporary AI bias works the same way — it's the emergent behavior of data pipelines, model architectures, deployment contexts, user interactions, feedback loops, organizational incentives, and regulatory environments all operating together.

This means that analyzing bias at any single layer will always give you an incomplete picture. You can audit the training data and find it "fair." You can test the model on benchmarks and find it "unbiased." You can review the deployment policy and find it "equitable." And the system can still produce systematically harmful outcomes — because the bias lives in the *interactions between layers*, not within any one of them.

MECHANISM: OPERATES THROUGH MEDIATION LAYERS IN COMPLEX

PATHWAYS

This is where the BBS model becomes not just useful but essential. In contemporary AI, information crosses multiple boundaries and transforms through multiple conceptual spaces before producing an outcome:

• A person exists in the phenomenological space (LP) with full human complexity — history, context, nuance, contradiction.

• That person encounters a boundary object — an interface, a form, a sensor, a data collection mechanism — that translates some subset of their reality into a computational representation.

• That representation enters the left virtual space (LV) where it's processed, combined with other data, transformed by algorithms, weighted by models, filtered by business logic.

• The output crosses another boundary into the right virtual space (RV) — a score, a recommendation, a classification, a decision — which then re-enters the phenomenological space as a real-world consequence.

At *every one of these transitions*, bias can enter, amplify, transform, or interact with bias from other transitions. A small encoding bias at the boundary object might be invisible in isolation but catastrophic when it interacts with a model architecture bias two layers deeper. This is why pathway analysis matters — you have to trace the full journey of information through the system to understand where the cumulative distortion comes from.

FRAMEWORK: PATHWAY-BASED ANALYSIS REVEALS INTERACTION PATTERNS

The key analytical move this slide introduces is shifting from *component-based* thinking ("is this dataset biased? is this model biased?") to *pathway-based* thinking ("how does information transform as it moves through the system, and where do those transformations produce or amplify systematic distortion?"). This is the methodological contribution of the BBS framework to AI ethics.

Pathway-based analysis asks different questions than component auditing:

• Instead of "is the data representative?" → "What was lost or distorted when reality was encoded into data, and how does that distortion interact with downstream processing?"

• Instead of "is the model fair?" → "How does the model's transformation of input representations into output representations systematically advantage or disadvantage certain groups, given the specific distortions already present in its inputs?"

• Instead of "is the outcome equitable?" → "Trace the full pathway from person to decision — at which boundary crossings did the cumulative distortion become consequential, and could intervention at an earlier crossing have prevented it?"

THE ACCOUNTABILITY CHALLENGE

If bias is distributed, responsibility must be too — but our current accountability structures aren't built for this. We want to assign blame to a person, a company, a dataset, or an algorithm. Distributed emergence resists this. The designer who chose the data schema, the team that collected the training data, the engineer who selected the model architecture, the product manager who set the optimization metric, the organization that deployed the system, the regulator that approved it — each contributed to the pathway, and the bias emerged from their collective contributions. This connects directly to slide 32 later in the deck on distributed responsibility.

TEACHING NOTE:

This slide is the theoretical capstone of the bias progression. Make sure students can articulate the difference between the four paradigms:

• Human bias (slide 12): individual, cognitive, correctable through awareness • Statistical bias (slide 13): systematic, formalized, correctable through better measurement — but also includes information loss, signal degradation, and conceptual space transformation that go beyond classical statistical correction • Early systems bias (slide 14): automated, explicit, correctable through code review • Modern ML bias (slide 15): learned, emergent, partially addressable through data and model interventions • Contemporary AI bias (slide 16): distributed, multi-layer, addressable only through pathway-based analysis

Each paradigm subsumes and complicates the previous one. Contemporary AI systems contain *all*of these bias types operating simultaneously — human bias in the designers, statistical bias in the data, rule-based bias in preprocessing, ML bias in the models, and emergent distributed bias in how they all interact. The BBS model gives students a framework for navigating this complexity rather than being overwhelmed by it.

CORE INSIGHT: MEDIATION IS CENTRAL

This slide is the conceptual hinge of the session. The prior slides (12–16) traced bias through increasingly complex system types — from human cognition to statistical measurement to hardcoded rules to ML to contemporary AI. The throughline across all of them is that bias is never just "in" a single component. It is *produced* through the mediating structures that connect components. This is the key move: from "where is the bias?" to "how does the system produce bias through its pathways?"

Mediation means that between any input and any output, there are intermediate processes that *transform* information. Each transformation is a potential site where bias enters, shifts, or amplifies. The input does not pass through the system unchanged — it is re-encoded, filtered, weighted, and recontextualized at every transition point. Understanding bias therefore requires understanding mediation: the structures, encodings, and boundary crossings that sit between the real world and the system's output.

FRAMEWORK: BBS MODEL

The BBS (Boundary-Based Systems) model provides the analytical framework for this mediation-centered view. Its core components map directly onto the bias analysis task:

• Boundary objects — The interfaces where information crosses from one representational context to another (e.g., a patient's symptoms become a diagnostic code, a job applicant's experience becomes a numerical score). Each crossing is a potential bias injection point because the translation always involves loss, simplification, or reframing.
• Virtual spaces — The representational contexts on either side of a boundary. Data means different things in different spaces: a "risk score" in the actuarial space is an abstract statistical construct, but in the lending decision space it determines whether someone gets a mortgage. The same number carries different implications depending on which virtual space it inhabits.
• Pathways — The sequences of boundary crossings that connect the life phenomenon (LP) to the system's output. Tracing a pathway means following how information transforms step by step, identifying where fidelity is lost, where assumptions are introduced, and where bias compounds.

The power of the BBS model is that it makes mediation visible and analyzable. Instead of asking "is this system biased?" (a binary question with limited usefulness), you can ask "at which boundary crossings does bias enter, and how does it transform as it moves through the pathway?" This gives you specific, actionable locations for intervention.

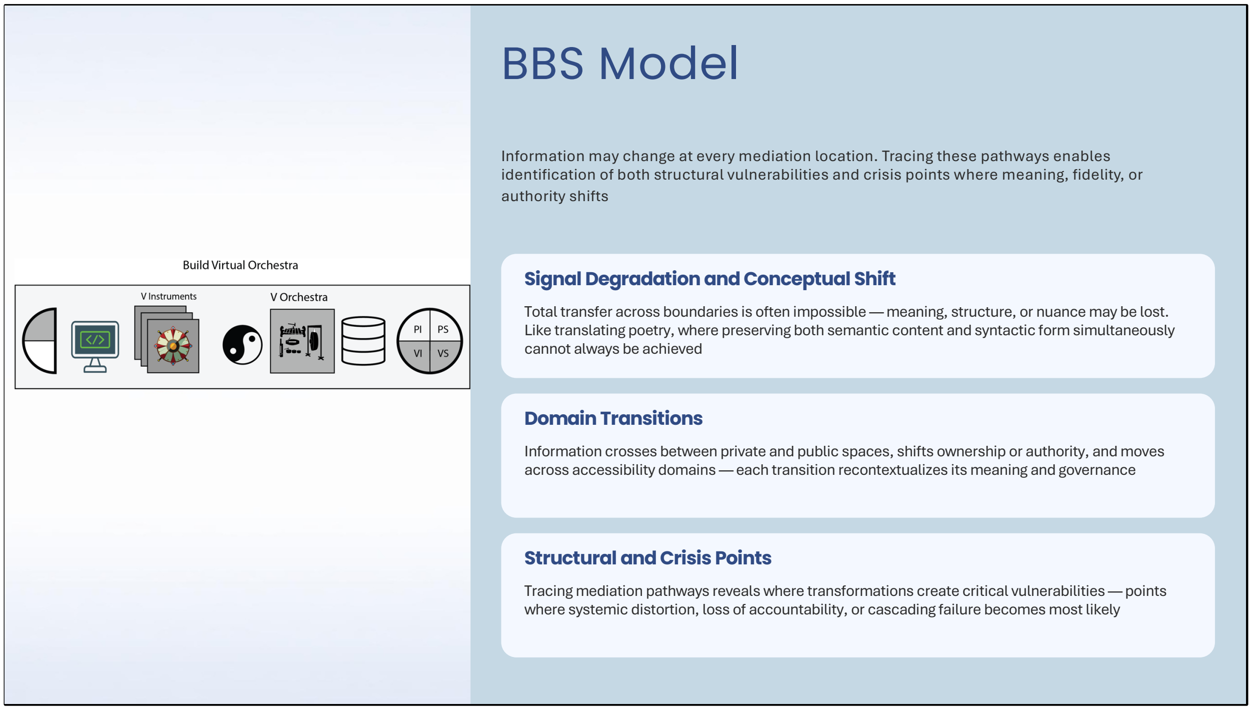
APPLICATION: SYSTEM ANALYSIS

Critically, pathway analysis is not limited to AI systems — it applies to any mediated process: bureaucratic workflows, legal systems, healthcare triage, supply chains, educational assessment. Anywhere information passes through mediating structures, the same analytical framework applies. And the analysis serves two distinct purposes: it reveals how subtle distortions at each transition can compound into systematic inequities, but it also identifies major structural bias sources — single boundary crossings or design choices that function as central points of distortion, dominating the system's overall bias profile. When students apply this to their Assignment 7 systems, they should be tracing pathways — not just naming components. The analysis requires identifying:
• Where does real-world complexity get reduced to system-readable data? (Boundary crossing: LP → LV)
• What assumptions are embedded in how data is encoded and processed? (LV transformations)
• Where do system outputs cross back into real-world consequences? (Boundary crossing: RV → RP)
• How do feedback loops create compound effects? (Pathway cycles)

The goal is not to find "the bias" as if it's a single defect. Pathway analysis operates at two scales simultaneously: it maps how small, individually defensible design choices at each transition compound into systematic patterns of advantage and disadvantage, and it identifies major structural bias sources — central points in the pathway where a single design decision, encoding choice, or institutional assumption introduces dominant distortion that shapes everything downstream. Some boundary crossings matter far more than others, and pathway analysis makes that hierarchy visible. This applies to all systems, not just AI — a hospital intake form, a credit scoring methodology, a university admissions rubric, or a zoning regulation each contain mediation pathways with identifiable central bias points. This is what pathway analysis reveals that component-level auditing misses.

TEACHING NOTE:

This slide synthesizes the entire bias progression (slides 12–16) into a single actionable principle. Use it as the pivot before moving to system-level topics. Have students revisit their Assignment 7 systems using the mediation lens — specifically, ask them to trace one complete pathway from real-world input to real-world consequence and identify every boundary crossing where bias could enter or transform. The most common student mistake is to identify bias as a property of a component (e.g., "the training data is biased") rather than as a property of the pathway (e.g., "bias enters when clinical observations are encoded as binary diagnostic codes, which erases the spectrum of severity that existed in the original patient encounter"). Push them toward the pathway framing.



SIGNAL DEGRADATION AND CONCEPTUAL SHIFT

The central insight here is that information does not pass through mediation locations unchanged. At every boundary crossing, the possibility of signal degradation or conceptual transformation exists. This is not a bug in the system — it is an inherent property of mediation itself. Whenever information moves from one representational context to another, something must be translated, and translation is never lossless.

The poetry translation analogy is powerful here. When you translate a poem from one language to another, you face a fundamental impossibility: you must preserve both the semantic content (what the poem means) and the syntactic structure (how it is constructed — meter, rhyme, rhythm, word order). These two dimensions are deeply entangled in the source language, and the target language has its own entanglements. Preserving both simultaneously is often impossible. You must choose which dimension to sacrifice, and that choice itself introduces a form of bias — it privileges one aspect of the original over another.

This is exactly what happens at boundary crossings in any mediated system — not only AI. A patient's lived experience of chronic pain gets encoded as a 1–10 numerical score: the semantic richness (what the pain means to their life) is sacrificed for syntactic compatibility (a number the system can process). A legal precedent gets summarized in a case digest: the nuance of judicial reasoning is compressed into a categorical holding. A student's creative essay gets reduced to a rubric score. In each case, the concept changes as it is transformed — it enters the boundary as one thing and exits as something related but fundamentally different.

The impossibility of total transfer is not a failure of implementation — it is a structural property of mediation. Every boundary crossing involves a choice about what to preserve and what to let go, and that choice has ethical implications even when it appears purely technical.

DOMAIN TRANSITIONS

Beyond signal fidelity, information also changes its governance and accessibility status as it moves through pathways. These domain transitions are a distinct category of transformation that often goes unexamined:

• Private → Public: A medical record shared with an insurer; a confidential employee evaluation aggregated into a departmental report; personal location data collected by an app and sold to advertisers. The information itself may not change, but its context shifts dramatically — what was private becomes visible, and the individual loses control over how it is interpreted.

• Ownership and authority shifts: Data collected under one authority (a hospital, a university, a government agency) gets transferred to another authority with different rules, incentives, and purposes. The same data now serves different masters. A student's educational record, created under FERPA protections, might feed into a workforce analytics platform with entirely different governance. The information crosses an authority boundary, and the rules that governed its use no longer apply.

• Accessibility domains: Information that was available to a narrow group becomes broadly accessible (or vice versa). A court record that was technically public but practically obscure becomes easily searchable when digitized. The accessibility transformation changes the effective meaning of the information even though the content remains identical.

These domain transitions are critical because they represent points where the ethical framework governing the information changes, often without the knowledge or consent of the people the information describes.

STRUCTURAL AND CRISIS POINTS

The purpose of tracing mediation pathways is not merely descriptive — it is diagnostic. By mapping where information transforms, where domains shift, and where signal degradation occurs, we can identify two categories of critical locations:

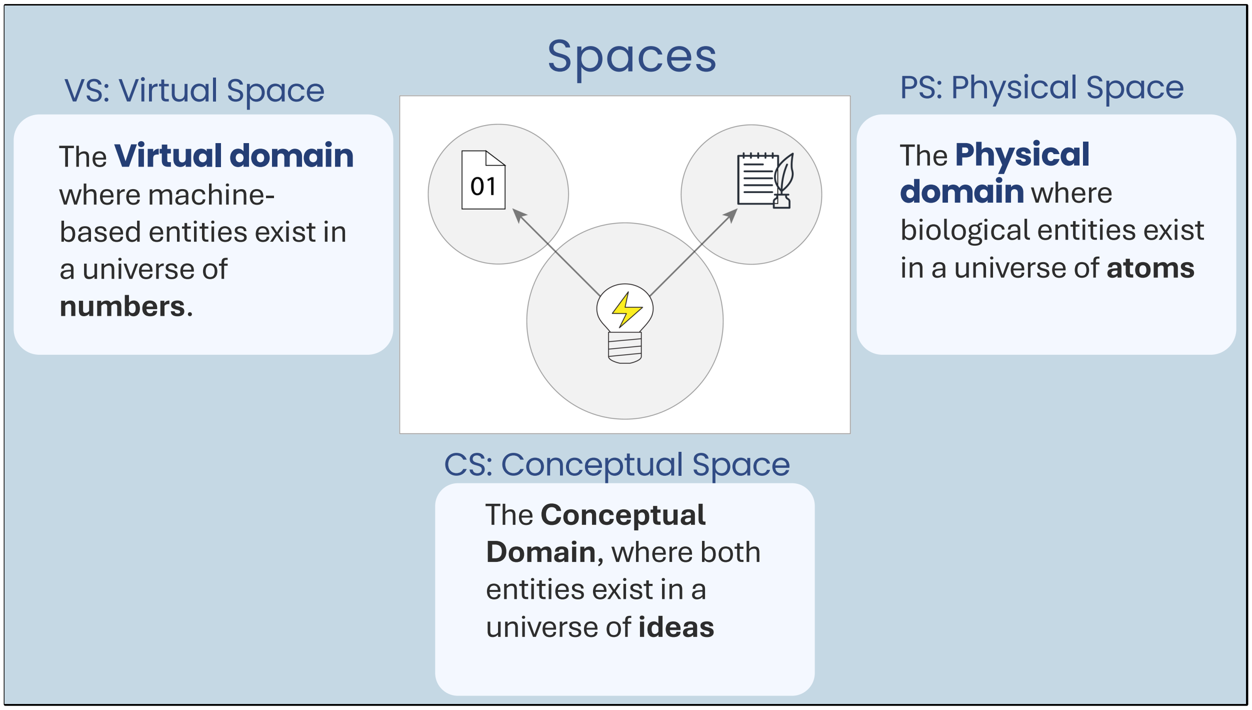
• Structural points — Boundary crossings where the system's architecture creates persistent, repeatable distortions. These are the locations that produce systematic bias through their normal operation. They don't require failure to produce harm — they produce it by design, even if that design was well-intentioned.

• Crisis points — Locations where transformations create cascading vulnerabilities. A single misclassification at an early boundary can propagate through the entire pathway, amplifying at each subsequent crossing. These are the points where system failures become catastrophic rather than contained — where a local error becomes a global distortion.

Identifying both categories requires pathway-level analysis. Component-level auditing can find individual errors but cannot reveal how those errors propagate, amplify, or interact across the system.

TEACHING NOTE:

The poetry translation analogy is useful as a quick illustration but does not need deep exploration — the point is simply that total transfer is impossible, and every boundary crossing involves trade-offs between dimensions that cannot all be preserved simultaneously. Use it as a doorway to the broader principle, not a destination. The domain transition examples (private/public, ownership, accessibility) are especially effective because students can connect them immediately to their own digital lives — they intuitively understand that their social media data being used by an advertiser feels different from sharing it with friends, even though the data is "the same." Use this intuition to build toward the more abstract BBS framing. Push students to identify at least one structural point and one crisis point in their Assignment 7 systems — this forces them to distinguish between persistent design-level bias and cascading failure modes, which are analytically and ethically distinct.



SPACES: THE THREE DOMAINS OF THE BBS MODEL

Purpose: This slide introduces the foundational ontology of the BBS (Boundary-Based Systems) model — the three spaces in which entities and information exist. Understanding these spaces is prerequisite to understanding how information transforms as it crosses boundaries between them.

PS: PHYSICAL SPACE

The Physical Space is the domain of atoms, biology, and material reality. This is where humans exist as embodied beings — with sensory experience, physical presence, and biological constraints. In the BBS model, Physical Space is where the "life phenomenon" (LP) originates: the actual person, situation, or event that the system will ultimately process.

Key properties of Physical Space:

• Continuous and analog — physical reality doesn't come in discrete categories

• Context-rich — every physical entity exists within a web of relationships, histories, and circumstances

• Directly experienceable — humans can perceive and interact with physical space through senses

• Inherently complex — no representation can fully capture physical reality

For bias analysis, Physical Space matters because this is where the full complexity of human experience exists before any system reduces it. A person's creditworthiness, health status, or job qualification exists in Physical Space as a rich, multidimensional, context-dependent reality. Everything that follows is a reduction.

VS: VIRTUAL SPACE

The Virtual Space is the domain of numbers, data structures, and computational representations. This is where machines operate — processing information according to rules, algorithms, and learned patterns. In the BBS model, Virtual Space is where information exists after it has been encoded and before it produces a real-world consequence.

Key properties of Virtual Space:

• Discrete and digital — everything is represented as numbers, categories, or symbols
• Context-stripped — the richness of physical reality is reduced to what the encoding captures
• Precisely manipulable — computational operations can transform virtual representations with mathematical exactness
• Semantically ambiguous — the same number can mean entirely different things depending on the virtual space it inhabits

For bias analysis, Virtual Space is where most transformation and amplification occurs. The encoding choices that define what enters Virtual Space, and the operations that transform information within it, are the primary mechanisms through which bias operates in computational systems.

CS: CONCEPTUAL SPACE

The Conceptual Space is the domain of ideas, meanings, interpretations, and frameworks. This is where both humans and machines exist when they are engaged in reasoning, categorization, and sense-making. Conceptual Space is shared between physical and virtual domains — it's where the meaning of information is constructed and contested.

Key properties of Conceptual Space:

• Interpretive — the same information can support different meanings depending on the conceptual framework applied
• Shared but contested — humans and systems may operate with different conceptual frameworks for the same domain
• Framework-dependent — what counts as "relevant," "similar," or "important" depends on the conceptual space in use

• Where bias becomes consequential — a biased representation only produces harm when it is interpreted and acted upon within a conceptual framework

For bias analysis, Conceptual Space is where we evaluate whether a system's operations are fair, appropriate, or harmful. The same computational output can be ethical in one conceptual framework and harmful in another. This is why bias analysis cannot be purely technical — it requires engagement with the conceptual frameworks that give system outputs their meaning.

THE THREE-SPACE INTERACTION
The critical insight is that information continuously moves between these spaces, and each crossing is a transformation. A patient's physical symptoms (PS) are encoded as diagnostic codes (VS) which are interpreted within a medical framework (CS). At each crossing, something is gained and something is lost. The BBS model provides tools for analyzing exactly what changes at each boundary — and where those changes introduce or amplify bias.

TEACHING NOTE: Students sometimes struggle with the abstraction level of this slide. Ground it immediately with a concrete example: "When you fill out a job application online, you exist in Physical Space (a complex person with a rich history). The application form is a boundary object that translates you into Virtual Space (a data record with fields and values). The hiring algorithm operates in Virtual Space to produce a score. That score is then interpreted in Conceptual Space (what does a '72' mean for this candidate?) and the decision crosses back into Physical Space when you receive a rejection email. Each of those crossings changed something about how 'you' were represented — and each is a potential site for bias."

Boundary Objects: Transduction

Devices and interfaces act as mediators with dual-state properties, existing in both physical and virtual forms

Mediation between spaces
Boundary objects connect and translate between different system domains

Dual-state objects
Exist simultaneously in both physical and virtual forms

Forms, sensors, APIs
Common examples of boundary objects in modern systems

BOUNDARY OBJECTS: TRANSDUCTION

Purpose: This slide introduces boundary objects — the devices, interfaces, and mediating structures that enable information to cross between spaces. The concept of transduction (converting information from one form to another) is central to understanding how bias enters and transforms within systems.

MEDIATION BETWEEN SPACES
Boundary objects are the interfaces where information crosses from one representational domain to another. They are not passive conduits — they are active transformers that shape what information can pass through and how it changes in transit. Every boundary object makes decisions (implicitly or explicitly) about what to capture, what to discard, what to categorize, and what format to impose.
Examples in everyday systems:
• A thermometer transduces temperature (physical) into a number (virtual)
• A web form transduces a person's self-description (physical/conceptual) into structured data fields (virtual)
• A microphone transduces sound waves (physical) into electrical signals and

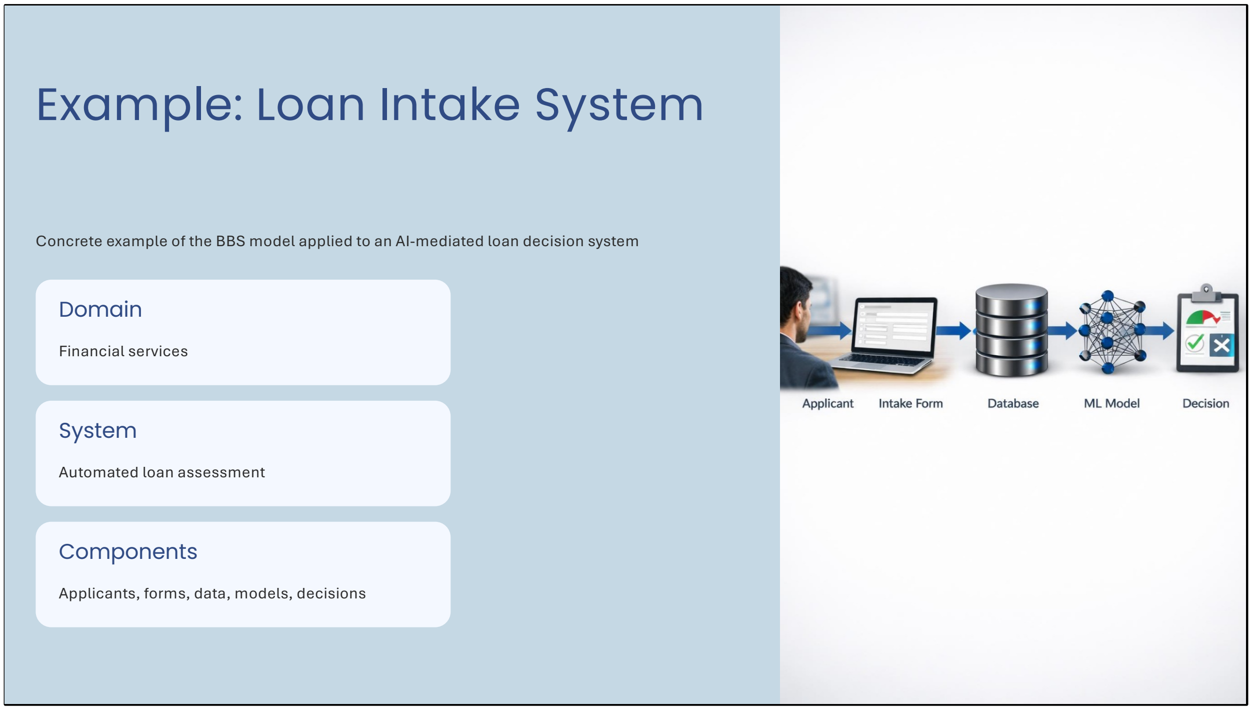
then digital samples (virtual)
• A diagnostic rubric transduces clinical observation (physical/conceptual) into a categorical score (virtual)

DUAL-STATE OBJECTS
The defining characteristic of boundary objects is that they exist simultaneously in two domains. A web form is both a physical interface (visible on a screen, interacted with by human hands) and a virtual structure (defining data types, field lengths, and validation rules). A sensor is both a physical device (located in space, subject to environmental conditions) and a virtual instrument (producing numerical outputs in defined formats).
This dual-state property is what makes boundary objects the critical sites for bias analysis:
• On the physical side, they determine what aspects of reality are accessible and how they are experienced
• On the virtual side, they determine what representations are created and what constraints are imposed
• The gap between these two sides — what the physical world offers and what the virtual representation captures — is where bias enters

FORMS, SENSORS, APIs
The slide lists three categories of boundary objects, each with distinct bias implications:
• Forms: Human-designed interfaces that impose categories, require specific information, and constrain self-expression. Every form field is a bias decision: what's included, what's excluded, what options are available, what's mandatory. The loan application form (slides 22–25) is the session's primary example.
• Sensors: Physical devices that transduce environmental phenomena into data. Sensor bias includes calibration assumptions (designed for certain conditions), sampling rates (what's captured vs. missed), and environmental factors (performance varies across contexts).
• APIs: Computational interfaces that define what data can be exchanged between systems and in what format. API bias includes data format constraints, required vs. optional fields, rate limits, and authentication requirements that determine who can access the system.

CONNECTION TO BIAS ANALYSIS
Every boundary object in a system is a potential bias entry point, transformation point, or amplification point. The analytical question is always: "What changes when information crosses this boundary? What is gained, lost, or distorted? And who is systematically advantaged or disadvantaged by those changes?"

TEACHING NOTE: The transduction concept is powerful because it reframes bias from a moral failing to a physical process. Just as a microphone can't capture all frequencies equally (it has a frequency response curve that amplifies some sounds and attenuates others), a web form can't capture all human experiences equally. The question isn't whether the boundary object introduces distortion — it always does — but whether that distortion systematically disadvantages certain groups. This is a more productive framing for students who resist moral language around bias.



EXAMPLE: LOAN INTAKE SYSTEM

Purpose: This slide introduces the concrete case study that will anchor the abstract BBS concepts in a real-world application. The loan intake system is chosen because it's familiar, consequential, and has well-documented historical bias issues.

DOMAIN: FINANCIAL SERVICES
The financial services domain is ideal for this example because:
• The stakes are high and tangible — loan decisions directly affect people's ability to buy homes, start businesses, and build wealth
• Historical bias is extensively documented — redlining, discriminatory lending practices, and algorithmic discrimination in credit scoring are well-studied
• The system is relatively simple to decompose — clear inputs (applicant data), clear processing (credit model), clear outputs (approve/deny)
• Students can relate to it — most have applied for credit, loans, or financial products

SYSTEM: AUTOMATED LOAN ASSESSMENT

The automated loan assessment system represents a complete mediated pathway from human applicant to institutional decision. It contains:

• Physical Space components: The actual human applicant with their full life context, financial history, and circumstances

• Boundary objects: Application forms, document upload interfaces, identity verification systems

• Virtual Space components: Data records, credit scores, risk models, decision algorithms

• Conceptual Space components: Definitions of "creditworthiness," "risk," "qualified borrower"

COMPONENTS: APPLICANTS, FORMS, DATA, MODELS, DECISIONS

Each component occupies a specific location in the BBS pathway:

• Applicants (LP — Life Phenomenon): The starting point. Real humans with complex, context-dependent financial situations that resist simple categorization.

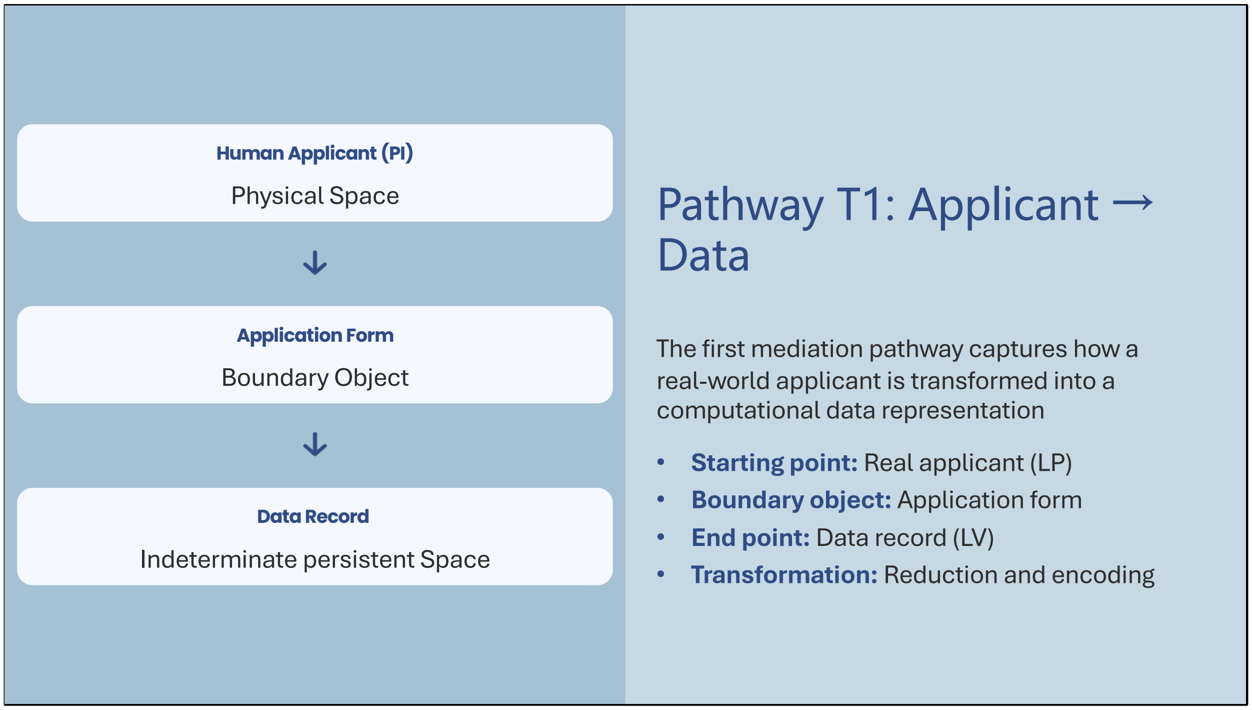
• Forms (Boundary Object): The first mediation point. Where the rich reality of a person's financial life is compressed into structured fields. This is where slides 24–25 will focus their bias analysis.

• Data (LV — Left Virtual Space): The encoded representation. The applicant now exists as a data record — a set of numbers and categories that the system can process.

• Models (Virtual Space Processing): The transformation engine. Algorithms that take data representations and produce risk assessments, scores, and recommendations.

• Decisions (RV — Right Virtual Space / RP — Right Phenomenon): The output that crosses back into the physical world as a real consequence for the applicant.

TEACHING NOTE: Frame this example as a pathway exercise, not just a system description. As you walk through slides 22–25, have students trace the full journey of an applicant through the system and identify every boundary crossing where information changes form. The goal is to build fluency with pathway analysis before they apply it to their own Assignment 7 systems. Emphasize that this is the simplest possible complete BBS pathway — real systems have far more boundary crossings and transformation steps.



PATHWAY T1: APPLICANT → DATA

Purpose: This slide maps the first mediation pathway in the loan system — the transformation of a real human applicant into a computational data record. This is the most consequential boundary crossing in the system because it determines what the system can "see" about the applicant.

STARTING POINT: REAL APPLICANT (LP)
The Life Phenomenon is the actual human being — with all their complexity, context, and dimensionality. Before encountering the system, the applicant exists as:
• A person with a financial history that includes context, circumstances, and narrative (not just numbers)
• Someone whose creditworthiness is a complex judgment that depends on factors the system may never capture
• A member of communities, families, and social networks that influence their financial situation
• An individual whose relationship to money and debt is shaped by cultural, historical, and personal factors

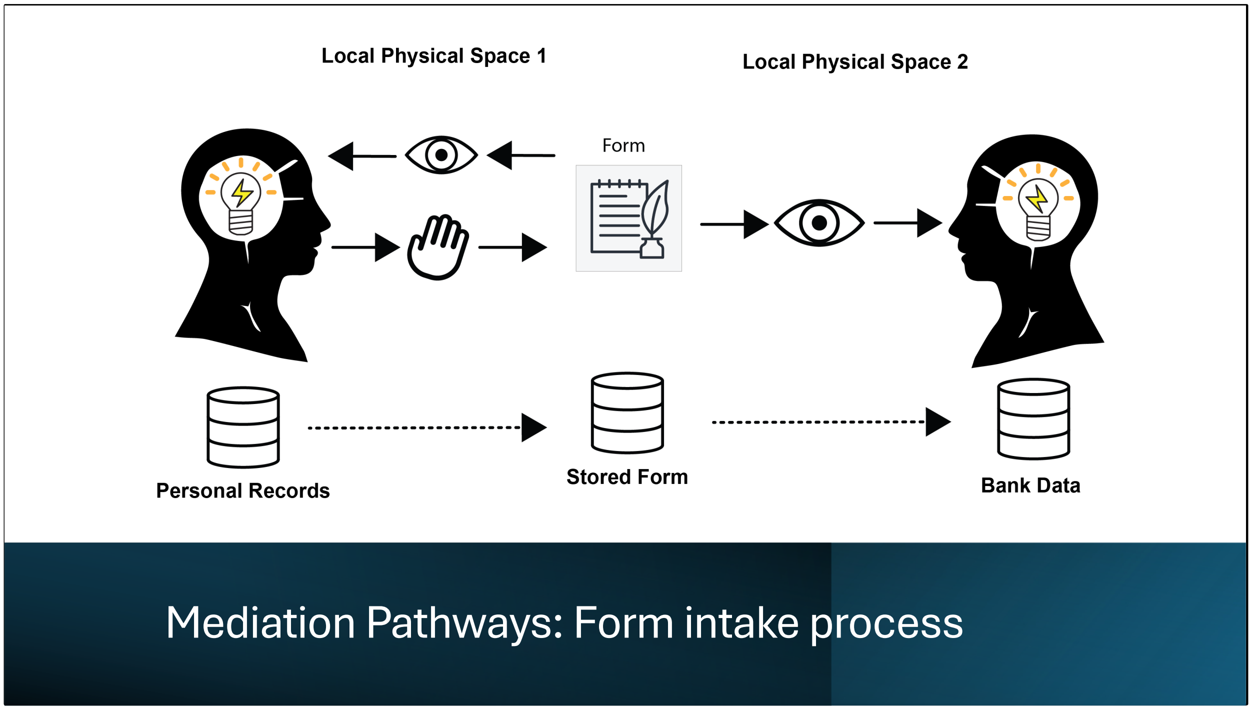
The key insight: everything the system will ultimately "know" about this person is constrained by what survives the boundary crossing at the application form. The LP contains infinitely more information than any data record can capture.

BOUNDARY OBJECT: APPLICATION FORM
The application form is the transduction device that converts a complex human into structured data. It is not a neutral mirror — it is an active filter that determines:
• What dimensions of the applicant are captured (income, employment, address) and which are invisible (caregiving responsibilities, informal economy participation, community ties)
• What categories are available (the dropdown options, the checkboxes, the field formats) and which life situations don't fit any category
• What precision is imposed (income as a single annual number vs. variable monthly amounts) and what nuance is lost
• What sequence and framing shapes responses (do you ask about assets before liabilities? do you ask about income before expenses?)

END POINT: DATA RECORD (LV)
After crossing the boundary, the applicant exists in Virtual Space as a data record — a structured collection of fields and values. This representation is:
• Dimensionally reduced — the hundreds of relevant aspects of a person's financial life are compressed into perhaps 20–50 fields
• Categorically constrained — continuous realities (income varies monthly) become discrete values (annual income: $52,000)
• Context-stripped — the reasons behind the numbers are not part of the record
• Apparently objective — the data record looks like a neutral description, but it's actually the product of dozens of design decisions about what to capture and how

TRANSFORMATION: REDUCTION AND ENCODING
The nature of this pathway crossing is reduction: the applicant loses dimensionality. But it's not random reduction — it's structured by the form's design choices, which reflect assumptions about what matters for loan decisions. Those assumptions are where bias enters. If the form was designed based on the financial patterns of one demographic group, it will capture that group's reality more accurately than others'.

TEACHING NOTE: This is the slide to make the BBS pathway analysis tangible. Walk through a specific example: "Imagine two applicants — one with a stable salaried job and one who is a freelance contractor with variable income. The form asks for 'Annual Income' as a single number. The salaried worker enters their exact salary. The freelancer has to estimate, average, or guess. The data record for both looks identical in structure, but the quality of representation is systematically different. That's bias introduced at the boundary object, and it will propagate through every subsequent stage of the system."



MEDIATION PATHWAYS: FORM INTAKE PROCESS

Purpose: This slide provides a visual representation of the form intake process as a mediation pathway. The diagram illustrates how information flows and transforms as it moves from the applicant through the form into the system.

VISUAL PATHWAY ANALYSIS
The diagram on this slide should be read as a pathway map — each element represents a stage in the mediation process, and the connections between them represent boundary crossings where information transforms.

Key points for discussion:
• The pathway is not a simple input-output pipe. Information doesn't just "flow through" — it is actively reshaped at every stage. The form intake process involves multiple sub-transformations: the applicant interprets the form's questions (conceptual space), translates their reality into the form's categories (boundary crossing), and the system validates, formats, and stores the result (virtual space processing).
• Each arrow in the diagram represents a potential bias site. Where information

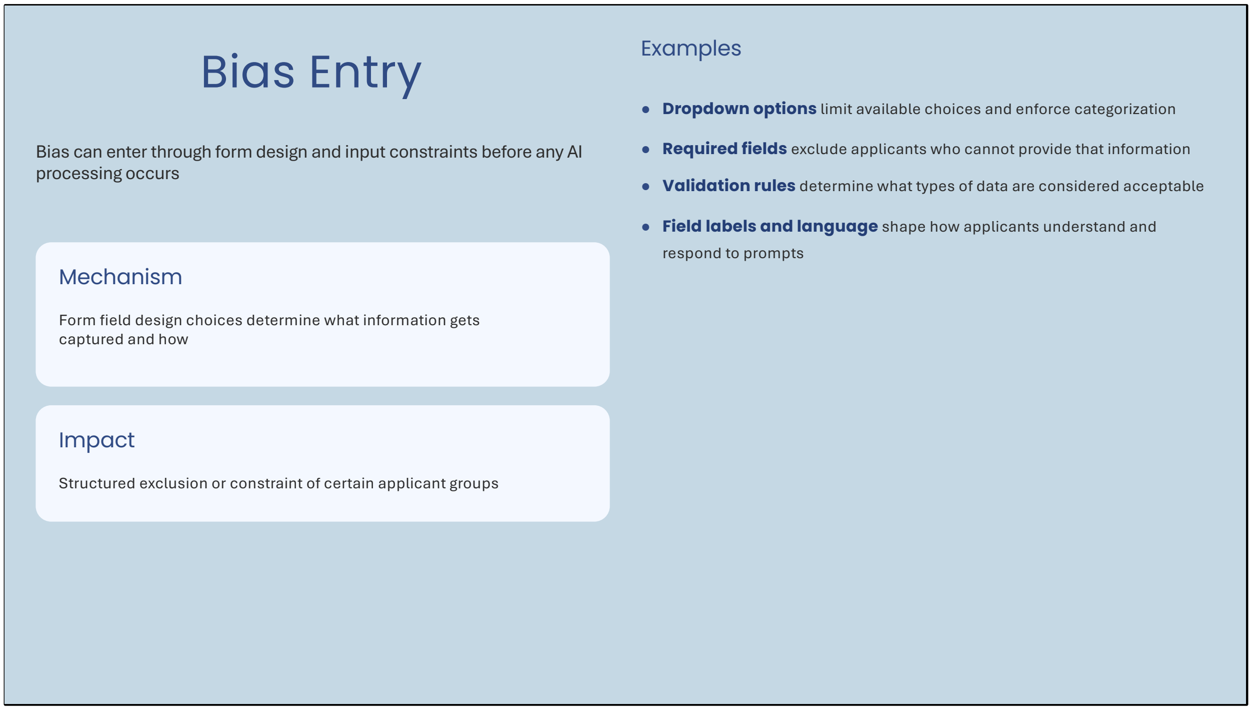
moves, it transforms. Where it transforms, assumptions are embedded. Where assumptions are embedded, systematic advantages and disadvantages can emerge.

• The visual representation helps students see that even a "simple" form intake involves multiple mediation steps. A single form field might involve: the applicant reading the label (conceptual interpretation), recalling relevant information (physical/cognitive), selecting or entering a value (physical/virtual boundary), the system validating the input (virtual processing), and storing it in a database format (virtual space encoding). Each step is a boundary crossing.

CONNECTION TO PRIOR SLIDES

This diagram concretizes what slide 18 described abstractly. The BBS model's concepts of boundary objects, virtual spaces, and pathway crossings are all visible here in a specific, traceable form. Use this visual as a reference point when students later analyze their own systems — they should be able to draw a similar pathway diagram for their Assignment 7 projects.

TEACHING NOTE: Ask students to identify at least three specific points in this diagram where bias could enter. Then ask: "If bias enters at the first point, how does it change by the time it reaches the third?" This builds the habit of thinking about bias as a pathway property rather than a point property. The form intake process is deceptively simple — it's actually a cascade of boundary crossings, each with its own bias implications.



EXAMPLES: BIAS ENTRY THROUGH FORM DESIGN

Purpose: This slide provides concrete, specific examples of how bias enters the loan system through form design choices. Each example illustrates a different mechanism of bias entry at the boundary object.

DROPDOWN OPTIONS LIMIT AVAILABLE CHOICES AND ENFORCE CATEGORIZATION

Expanded explanation: Every dropdown menu is a closed set of categories that someone decided was "complete enough." But real life doesn't come in dropdown options. Consider:

• Employment type: "Full-time / Part-time / Self-employed / Unemployed" — where does gig economy work go? Multiple part-time jobs? Seasonal employment? Informal caregiving economy? Each omitted category forces someone into a box that misrepresents their situation.

• Housing status: "Own / Rent / Other" — what about living with family (common in many cultures), co-op housing, employer-provided housing, or transitional situations? The categories reflect assumptions about what "normal" housing looks like.

• The bias isn't just in what's listed — it's in the cognitive effect on the applicant. When your real situation doesn't match any option, you're forced to pick the "closest" one, which introduces a systematic error that the system treats as accurate data.

REQUIRED FIELDS EXCLUDE APPLICANTS WHO CANNOT PROVIDE THAT INFORMATION

Expanded explanation: Marking a field "required" means that anyone who cannot provide that information cannot complete the application. This seems neutral but systematically excludes:

• People without traditional banking (requiring bank account numbers excludes the unbanked)

• People without stable addresses (requiring a fixed address excludes mobile or transitional populations)

• People without conventional employment documentation (requiring pay stubs excludes informal economy workers)

• Immigrants or refugees who may not have full documentation in the expected format

The requirement itself is the bias — it defines who can enter the system and who is excluded before any algorithm runs.

VALIDATION RULES DETERMINE WHAT TYPES OF DATA ARE CONSIDERED ACCEPTABLE

Expanded explanation: Validation rules are invisible gatekeepers. A phone number format that requires a US area code excludes international applicants. An income field that doesn't accept decimal points or ranges forces rounding. A name field that rejects special characters (accents, hyphens, apostrophes) literally cannot represent certain people's identities. These are technical implementation decisions that have demographic consequences.

FIELD LABELS AND LANGUAGE SHAPE RESPONSES

Expanded explanation: The words used on a form influence how people interpret and respond to questions. "Annual household income" vs. "your income" captures different realities. "Primary residence" assumes a single stable home. Medical forms that ask about "spouse" vs. "partner" signal assumptions about family structure. The language isn't just communication — it's a conceptual framework that advantages people whose lives match its assumptions.

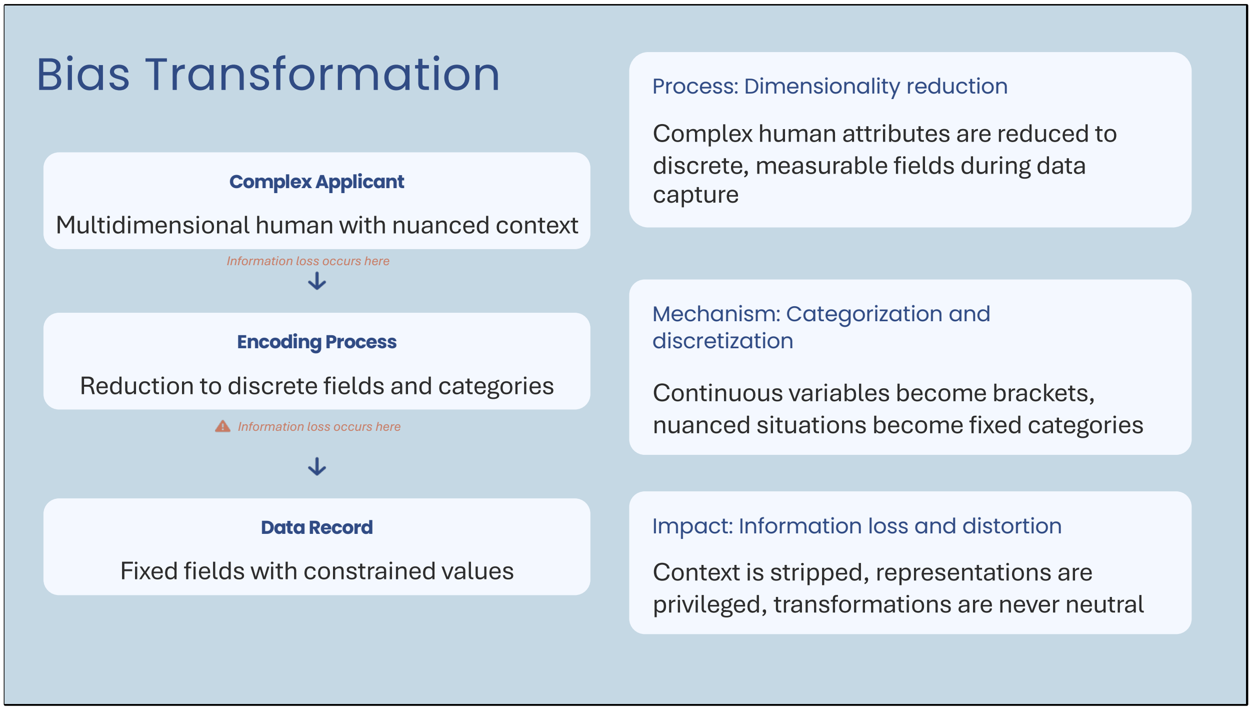
MECHANISM: FORM FIELD DESIGN CHOICES

The overarching mechanism is that form design is never neutral. Every field, every option, every validation rule, and every label encodes assumptions about what a "normal" applicant looks like. Those assumptions systematically advantage people who fit the assumed norm and disadvantage those who don't.

IMPACT: STRUCTURED EXCLUSION

The cumulative effect is not random error but structured exclusion — systematic patterns of who can successfully navigate the form and who cannot, who is accurately represented and who is distorted, who is included and who is filtered out before the system even begins processing.

TEACHING NOTE: These examples are deliberately drawn from the loan domain, but the principles apply universally. Ask students to identify analogous bias entry points in their Assignment 7 systems. Every system has forms, inputs, and interfaces — and every one of them makes these same types of design decisions with demographic consequences.



BIAS TRANSFORMATION: DIMENSIONALITY REDUCTION AND ENCODING

Purpose: This slide illustrates the core transformation that occurs when a complex human applicant is reduced to a data record. It makes visible the process of dimensionality reduction and encoding — the mechanism through which rich, contextual human reality becomes structured, context-stripped computational data.

PROCESS: DIMENSIONALITY REDUCTION
Dimensionality reduction is not just a technical operation — it is the fundamental act of mediation in any computational system. A human being exists in potentially infinite dimensions (every aspect of their experience, history, context, and circumstance). A data record exists in a fixed number of dimensions (the fields defined by the form and database schema). The transition from the former to the latter necessarily discards information.
The critical question is: what gets discarded, and who decides?

• The dimensions that survive are those that the system designers considered relevant, measurable, and encodable
• The dimensions that are lost are those that didn't fit the form's structure,

weren't considered important, or couldn't be easily quantified
• The choice of which dimensions to keep is itself a bias-generating decision, because it reflects a particular understanding of what matters — an understanding shaped by the designers' experience, training, and assumptions

COMPLEX APPLICANT → ENCODING PROCESS → DATA RECORD
The three-stage visual on this slide should be read as a progressive loss of information:
• Complex Applicant: A multidimensional human with nuanced context — financial history that includes reasons and circumstances, employment that includes trajectory and aspirations, housing that includes community and stability
• Encoding Process: The mechanism of reduction — continuous variables become brackets (income: $40,000–$60,000), nuanced situations become fixed categories (employment: "Self-employed"), contextual information becomes absent. "Information loss occurs here" is the key annotation.
• Data Record: The output of encoding — fixed fields with constrained values, context stripped, representations privileged, transformations invisible

MECHANISM: CATEGORIZATION AND DISCRETIZATION
Two specific mechanisms deserve attention:
• Categorization: Converting continuous or complex realities into discrete categories. "Employment type" forces a spectrum of work arrangements into 4–5 categories. "Marital status" forces complex relationship situations into a few checkboxes. Each categorization decision creates winners (people whose reality maps cleanly onto a category) and losers (people who must be shoehorned into the closest option).
• Discretization: Converting continuous values into ranges or brackets. Income brackets ($0–$25K, $25K–$50K, etc.) treat everyone within a range as identical. This erases meaningful differences within brackets and creates artificial cliffs between them. A person earning $49,999 and a person earning $50,001 may be treated very differently despite nearly identical incomes.

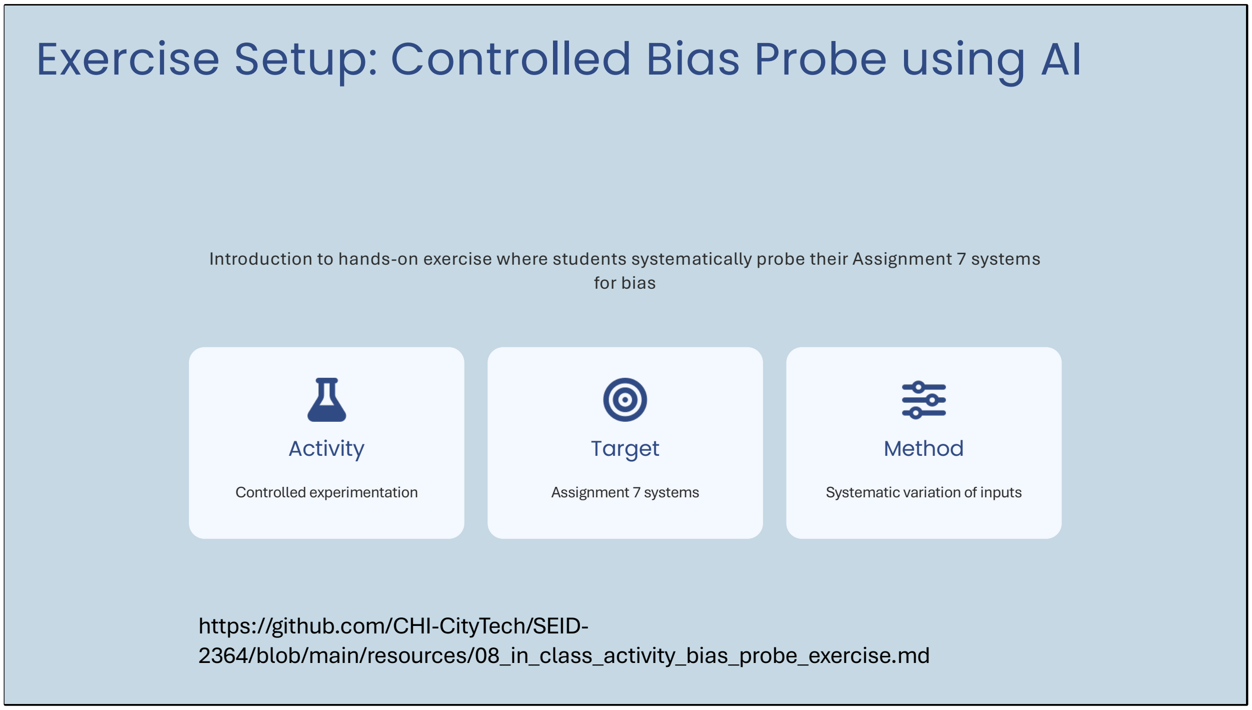
IMPACT: INFORMATION LOSS AND DISTORTION
The slide's three impact statements are each critical:
• "Context is stripped" — The reasons behind numbers are lost. A low income due to choosing part-time work for caregiving is treated the same as low income due to inability to find employment.
• "Representations are privileged" — People whose lives match the form's categories are represented more accurately than those whose lives don't. This is a systematic advantage.
• "Transformations are never neutral" — Every encoding choice benefits

someone and disadvantages someone else. The question is whether those benefits and disadvantages correlate with demographic characteristics.

CONNECTION TO BBS MODEL

In BBS terms, this entire slide illustrates a single boundary crossing: LP (complex applicant) → Boundary Object (form/encoding) → LV (data record). But within this single crossing, multiple sub-transformations occur, each with its own bias implications. This is why boundary analysis needs to be fine-grained — even a single boundary crossing contains layers of transformation.

TEACHING NOTE: The visual progression on this slide (complex person → encoding → data record) is one of the most important conceptual images in the session. Have students sketch an equivalent diagram for their Assignment 7 system: what is the "complex entity" that enters their system? What does the encoding process look like? What does the data record contain — and what was lost? This exercise builds the analytical muscle for the bias probe exercise on slides 26–29.



EXERCISE SETUP: CONTROLLED BIAS PROBE

Purpose: This slide sets up the hands-on exercise where students will systematically probe their Assignment 7 systems for bias. The exercise moves from theoretical understanding to empirical investigation.

ACTIVITY: CONTROLLED EXPERIMENTATION
The word "controlled" is deliberate and important. This is not casual exploration — it's systematic investigation with a method. The scientific approach matters because:
• Casual testing often confirms what you already expect rather than revealing what you don't
• Systematic variation isolates which variables produce differential treatment
• Controlled conditions allow you to attribute output changes to specific input changes rather than noise or randomness
• Documenting the method makes findings reproducible and comparable across students

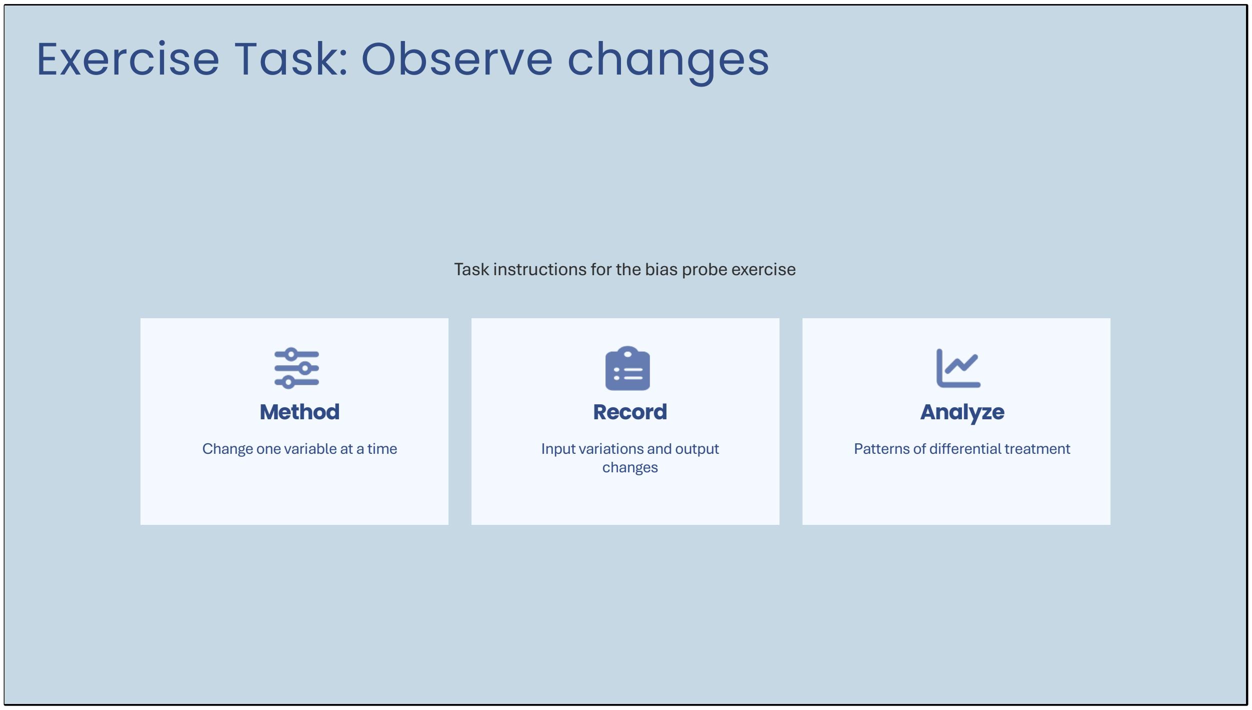
TARGET: ASSIGNMENT 7 SYSTEMS

Students are probing their own systems, which serves multiple pedagogical purposes:
• They already understand the system's intended purpose and design
• They have a personal stake in the findings (it's their work being analyzed)
• They can connect abstract bias concepts to concrete system behaviors they've built
• Discovering bias in your own system is more impactful than reading about bias in someone else's

METHOD: SYSTEMATIC VARIATION OF INPUTS
The method is input variation with output observation — the simplest form of bias testing:
• Change one variable at a time while holding others constant
• Choose variables that might serve as demographic proxies (names, locations, language patterns)
• Record both the input change and the output change precisely
• Look for patterns across multiple variations, not just individual differences

PREPARATION GUIDANCE FOR STUDENTS:
Before beginning the exercise, students should:
1. Identify 3–5 input variables to test (e.g., applicant name, location, employment description, writing style)
2. Prepare a baseline input that represents a "standard" or "neutral" case
3. Plan specific variations for each variable (e.g., names from different cultural backgrounds, addresses from different neighborhoods)
4. Set up a recording format (table or spreadsheet) with columns for: variable changed, original value, new value, output change observed, magnitude of change

TEACHING NOTE: Emphasize that the goal is not to "prove" their system is biased or "prove" it's fair. The goal is to observe systematically and document what they find. Some systems will show clear differential treatment; others may not with the variables tested. Both results are informative. If a system doesn't show bias with the variables tested, that doesn't mean it's bias-free — it means you need to test different variables or examine the pathway more deeply.



EXERCISE TASK: OBSERVE CHANGES

Purpose: This slide provides the specific procedural instructions for conducting the bias probe. The three steps — method, record, analyze — structure the exercise into a rigorous investigative process.

METHOD: CHANGE ONE VARIABLE AT A TIME
This is the core methodological principle. Changing multiple variables simultaneously makes it impossible to attribute output differences to any specific input change. The "one variable at a time" approach:
• Provides causal clarity: if you change only the name and the output changes, you know the name influenced the output
• Builds a systematic evidence base: each single-variable test adds one data point to your bias map
• Reveals proxy effects: sometimes changing a "non-sensitive" variable (like zipcode) produces the same differential treatment as changing a "sensitive" variable (like race), revealing that the system uses proxies

Specific testing suggestions:

• Names: Try names associated with different ethnic, cultural, or gender identities while keeping all other information identical
• Locations: Try addresses from different neighborhoods, cities, or regions with different demographic profiles
• Language: Try the same content written in different styles, registers, or with different cultural markers
• Edge cases: Try inputs at the boundaries of categories (e.g., income just above vs. just below a round number)

RECORD: INPUT VARIATIONS AND OUTPUT CHANGES
Documentation is essential because:
• Memory is unreliable for detecting subtle patterns across multiple tests
• Written records allow comparison across students during the synthesis phase (slides 30–31)
• The recording format itself forces precision — "the output seemed different" becomes "the confidence score dropped from 0.87 to 0.71 when the name was changed from John to Jamal"
• Records create evidence that can be analyzed for patterns rather than individual anecdotes

Recording format should capture:
• Exact input used (copy-paste, not paraphrased)
• Exact output received (copy-paste, not summarized)
• What specifically changed between the baseline and the test case
• Magnitude and direction of any output change

ANALYZE: PATTERNS OF DIFFERENTIAL TREATMENT
The analysis step is where individual observations become systemic findings:
• Look for consistency: Does the same type of change produce the same direction of output change across multiple tests?
• Look for magnitude: Are some variables more influential than others? This reveals the system's implicit hierarchy of what "matters"
• Look for proxies: Do seemingly neutral variables (zipcode, school name, employer name) produce differential treatment that correlates with demographic patterns?
• Look for thresholds: Are there input values where small changes produce large output jumps? These threshold effects are amplification points.

TEACHING NOTE: Circulate during this exercise and check that students are being systematic rather than random. The most common mistake is testing too many variables too quickly without recording carefully. Better to test three variables thoroughly than ten variables sloppily. Also watch for students who test only "obvious" bias vectors (name, gender) — encourage them to test structural variables (input format, level of detail, order of information) that reveal more subtle pathway biases.

WORK TIME: RUN EXPERIMENT

Purpose: This is the active experimentation phase where students conduct their controlled bias probes. The instructor's role shifts from lecturing to facilitating and coaching.

DURATION
The time allocation for this exercise should balance depth with practicality:
• Minimum 15 minutes for meaningful experimentation (less than this produces superficial tests)
• Recommended 20–25 minutes for thorough investigation
• Maximum 30 minutes before diminishing returns set in
• Give a 5-minute warning so students can complete their current test and document findings

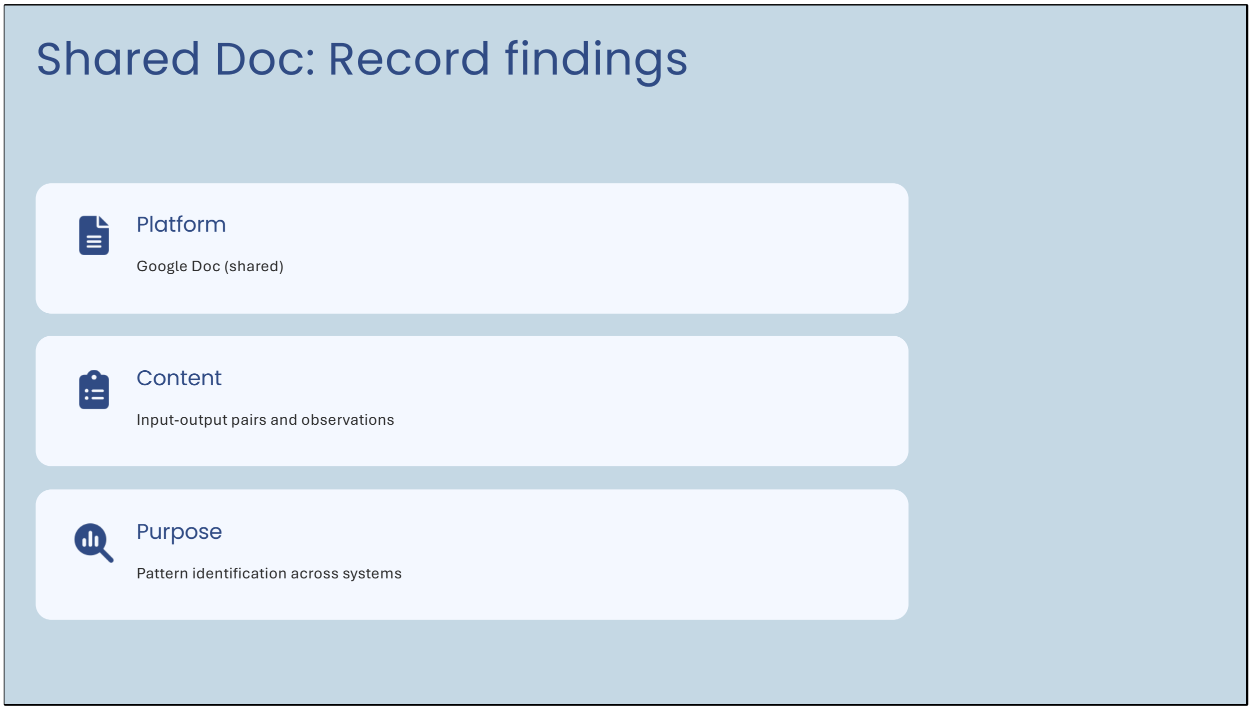
ACTIVITY: CONDUCT CONTROLLED BIAS PROBE
During the work time, students should be:
1. Running their baseline test case through their Assignment 7 system and recording the output

2. Systematically varying one input variable at a time and recording each output
3. Noting not just whether the output changed, but how it changed (direction, magnitude, nature)
4. Testing at least 2–3 different types of variables (e.g., one name-based, one location-based, one content-based)
5. Recording everything in their documentation format

INSTRUCTOR FACILITATION DURING WORK TIME:
• Circulate and check recording quality — students often observe differences but fail to document them precisely
• Prompt deeper investigation: "You found the output changed when you changed the name. What happens if you change it to a different name from the same cultural group? Is the effect consistent?"
• Help students who are stuck: some systems may not show obvious differential treatment with simple name or location changes. Suggest testing: input length, level of detail provided, order of information, presence/absence of optional fields, formal vs. informal language
• Watch for students who conclude "no bias found" too quickly — the absence of detectable bias with the variables tested is itself a finding worth recording, but shouldn't end the investigation
• Note interesting findings for the synthesis discussion (slides 30–31) — identify 3–4 students with particularly illustrative results to share

RESOURCES: ASSIGNMENT 7 SYSTEMS
Students should have their Assignment 7 systems readily accessible. If some students' systems require setup time or have rate limits, pair them with other students for the exercise rather than losing experimentation time to technical issues.

TEACHING NOTE: This is one of the most impactful exercises in the course because students are discovering bias in their own work. Be prepared for a range of emotional responses — some students will be surprised and concerned, others may be defensive ("my system isn't biased, you just asked it a weird question"). Both responses are pedagogically productive. The surprised students have had a genuine learning moment. The defensive students need help distinguishing between "the system is biased" (a moral judgment they resist) and "the system produces differential treatment for inputs that differ only on this dimension" (an empirical observation they can accept).



SHARED DOC: RECORD FINDINGS

Purpose: This slide transitions from individual experimentation to collective documentation. The shared document serves as both a data collection instrument and a collaboration platform for the synthesis phase.

PLATFORM: GOOGLE DOC (SHARED)
The shared document format is chosen because:
• Real-time collaboration allows students to see others' findings as they're posted
• Shared visibility reveals patterns that no individual student might notice on their own
• The collective document creates a class-level dataset for analysis
• It provides a permanent record that students can reference in their final projects

Document structure recommendation:
• One section per student (identified by name or student ID)
• Within each section: system description, variables tested, input-output pairs,

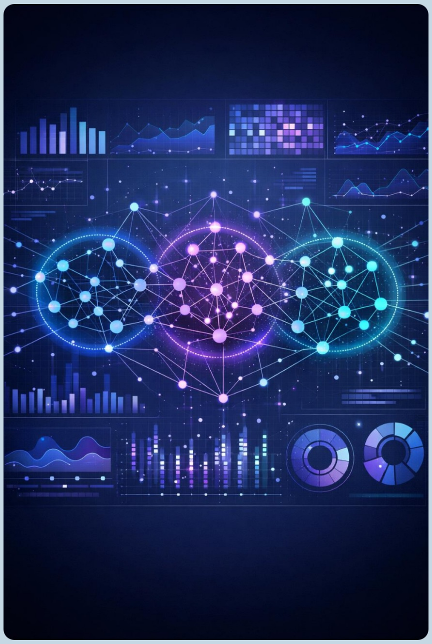
observed patterns
• A synthesis section at the bottom for cross-system patterns (to be filled during slide 30 discussion)

CONTENT: INPUT-OUTPUT PAIRS AND OBSERVATIONS
What students should record:
• Their system's name and brief description (what it does, what inputs it takes)
• The baseline input used and its output
• Each variation tested: what was changed, what the new output was, how it differed from baseline
• Qualitative observations: "The system seemed more confident when..." or "The language of the output shifted when..."
• Any patterns they noticed across multiple tests

Quality guidance:
• Exact inputs and outputs (copy-paste), not paraphrases
• Specific, measurable differences ("score dropped from 87 to 71") rather than vague observations ("output was worse")
• Clear identification of which variable was changed in each test
• Honest reporting of null results (tests where no difference was observed)

PURPOSE: PATTERN IDENTIFICATION ACROSS SYSTEMS
The collective documentation serves the synthesis phase (slides 30–31) by enabling:
• Cross-system pattern recognition: Do different types of AI systems show similar bias patterns?
• Variable sensitivity comparison: Which input variables produce the most consistent differential treatment across systems?
• Mechanism identification: Can students identify common mechanisms (proxy variables, threshold effects, category biases) across different system types?
• Evidence base for the "bias is mediated" insight (slide 31): The collective data should demonstrate that bias operates through system pathways rather than residing in any single component

TEACHING NOTE: Give students 5–7 minutes to transfer their findings to the shared document. Some will have extensive notes; others will have minimal documentation. Encourage the latter group to add qualitative observations even if their quantitative data is thin. During the synthesis discussion, you'll draw on the shared document as a live reference — scroll through it, highlight interesting findings, and draw connections between different students' observations. The document becomes a teaching tool for the remainder of the session.



Synthesis: Patterns of bias

- **Cross-system comparison** reveals common structural patterns across diverse AI applications
- **Name-based proxies** and address discrimination emerge as recurring vulnerabilities in pathway design
- **Decision thresholds** show differential treatment patterns that amplify initial biases
- Patterns reflect **fundamental challenges** in mediation structure, not isolated bugs

SYNTHESIS: PATTERNS OF BIAS

Purpose: This slide facilitates the instructor-led synthesis of the bias probe exercise. It draws cross-system patterns from students' collective findings and connects them to the session's theoretical framework.

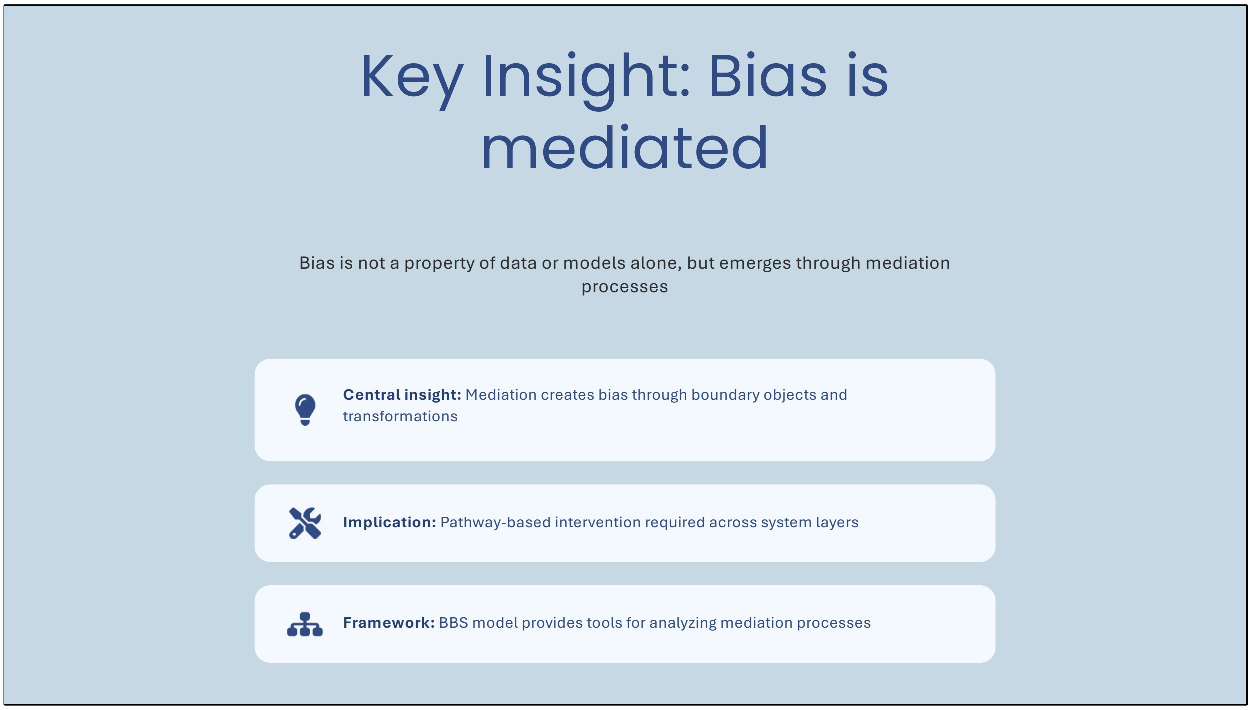
CROSS-SYSTEM COMPARISON REVEALS COMMON STRUCTURAL PATTERNS
The synthesis should demonstrate that bias isn't unique to any particular system — it manifests through common structural mechanisms across diverse applications. Guide the discussion to surface:
• Which variables consistently produced differential treatment across multiple students' systems? (Names and location tend to be the most reliable bias vectors)
• Were the same types of bias mechanisms present in different system domains? (e.g., proxy effects appeared in both the healthcare-focused and the hiring-focused systems)
• Did the magnitude of differential treatment vary predictably by system type or domain?

NAME-BASED PROXIES AND ADDRESS DISCRIMINATION
These are highlighted because they are among the most common and most well-documented forms of AI bias:
• Name-based proxies: Names carry demographic information (ethnicity, gender, age, socioeconomic background). When systems produce different outputs for identical inputs that differ only in name, it reveals that the system is using name as a proxy for demographic characteristics — even if no one programmed it to do so. This is emergent bias (slide 15) operating through learned correlations.
• Address discrimination: Geographic information is one of the strongest proxies for race and socioeconomic status due to the history of residential segregation. Systems that produce different outputs based on address or zipcode may be functionally reproducing redlining patterns through computational means. This connects directly to the early systems bias discussion (slide 14) and shows that historical bias patterns persist in modern systems.

DECISION THRESHOLDS SHOW DIFFERENTIAL TREATMENT
Threshold effects are particularly important for the amplification concept:
• When a continuous score is converted to a binary decision (approve/deny, flag/pass), the threshold placement determines who is included and who is excluded
• If the underlying score is systematically different for different groups (due to upstream bias), the threshold amplifies that difference into a categorical outcome
• Students may have observed this in their experiments: small input changes sometimes produced no output change (both cases fell on the same side of a threshold) and sometimes produced dramatic output changes (cases fell on opposite sides)

PATTERNS REFLECT FUNDAMENTAL CHALLENGES IN MEDIATION STRUCTURE
The key synthesis point: the patterns students observed are not bugs in specific systems. They are structural properties of how all mediated systems operate:
• Every system that encodes names creates the possibility of name-based proxy effects
• Every system that uses geographic data creates the possibility of address-based discrimination
• Every system that applies thresholds creates the possibility of amplified differential treatment
These are properties of the mediation pathway, not of any particular implementation. This is the empirical evidence for the theoretical claim that "bias is mediated" (slide 31).

TEACHING NOTE: The synthesis is the most important part of the exercise sequence. Students did individual experiments; now you need to help them see the collective pattern. Walk through the shared document on screen, highlighting 3–4 examples that illustrate different bias mechanisms. Then step back and ask: "What do all of these have in common?" Guide them toward the answer: they all involve bias entering, transforming, or amplifying through the system's mediation pathways. This sets up the "Key Insight" on slide 31.



KEY INSIGHT: BIAS IS MEDIATED

Purpose: This slide crystallizes the session's central theoretical contribution. After building through definitions (slides 12–16), framework (slides 17–18), spatial model (slide 19), boundary analysis (slides 20–22), concrete example (slides 23–25), and empirical exercise (slides 26–30), students should now be able to understand and defend this claim.

CENTRAL INSIGHT: MEDIATION CREATES BIAS THROUGH BOUNDARY OBJECTS AND TRANSFORMATIONS
This is the single most important sentence in the session. Unpack it carefully:
• "Mediation creates bias" — Bias is not something that pre-exists in data or algorithms and merely passes through systems. Systems actively produce bias through their mediation structures. The boundary crossings, encodings, transformations, and decision mechanisms don't just transmit bias — they generate it.
• "Through boundary objects" — The specific sites of bias production are the interfaces where information crosses between representational domains. The form, the sensor, the API, the encoding scheme, the threshold — each is a

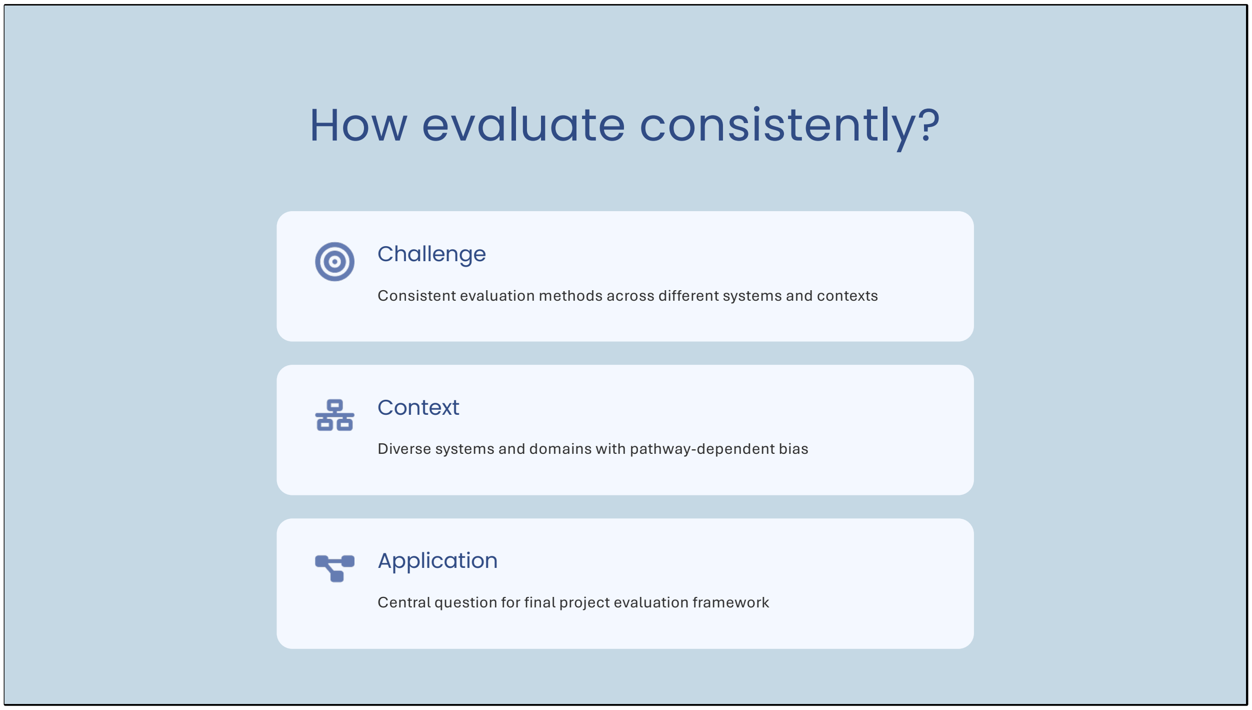
boundary object where bias can enter, transform, or amplify.
• "And transformations" — Bias is not static. It changes form as it moves through pathways. An input bias (limited dropdown categories) becomes a processing bias (proxy variable effects) becomes an output bias (differential approval rates). Tracing these transformations is the analytical contribution of the BBS model.

IMPLICATION: PATHWAY-BASED INTERVENTION REQUIRED ACROSS SYSTEM LAYERS
If bias is produced through pathways rather than residing in components, then intervention must also be pathway-based:
• Fixing the data alone won't fix the system if the encoding, processing, and threshold stages introduce new bias
• Fixing the model alone won't fix the system if biased data enters through the boundary objects
• Fixing the output alone (e.g., post-hoc fairness adjustments) won't fix the system if the underlying pathway continues to produce distortion
• Effective intervention requires identifying which boundary crossings produce the most consequential bias and intervening at those specific points

This doesn't mean you need to fix everything simultaneously. Pathway analysis reveals which boundary crossings are the most influential — the structural points where the most significant bias enters or amplifies. Targeted intervention at these critical points can have outsized impact on system-level fairness.

FRAMEWORK: BBS MODEL PROVIDES TOOLS FOR ANALYZING MEDIATION PROCESSES
The BBS model is not just a descriptive framework — it's an analytical toolkit:
• Identify all boundary crossings in the system pathway
• At each crossing, analyze what information is gained, lost, or distorted
• Determine whether those gains, losses, or distortions systematically advantage or disadvantage specific groups
• Prioritize interventions at crossings with the highest bias impact
• Verify that interventions at one crossing don't introduce new bias at subsequent crossings

TEACHING NOTE: This slide should feel like an arrival, not an introduction. If the session has worked as intended, students should be able to articulate this insight themselves before you state it. Consider asking: "Based on everything we've discussed and everything you observed in the exercise, if you had to state one principle about how bias operates in AI systems, what would it be?" Let students approximate the insight, then use this slide to formalize and sharpen their articulation. The power of the insight is in the journey to it, not in the statement itself.



HOW DO WE EVALUATE CONSISTENTLY?

Purpose: This slide pivots from analysis to methodology. Having established that bias is mediated and operates through pathways, the question becomes: how do we evaluate systems for bias in a way that is consistent, repeatable, and transferable across different systems, domains, and evaluators?

CHALLENGE: CONSISTENT EVALUATION METHODS ACROSS DIFFERENT SYSTEMS AND CONTEXTS
The consistency challenge has several dimensions:
• Systems differ in architecture, domain, and complexity — a bias evaluation method for a loan system may not directly apply to a healthcare triage system or a content moderation system
• Evaluators differ in training, perspective, and priorities — a data scientist, a civil rights advocate, and a system designer may evaluate the same system differently
• Contexts differ in stakes, populations, and values — what counts as "acceptable" bias may vary across domains and cultural contexts
• Time creates inconsistency — systems evolve, populations shift, and evaluation criteria change

Why consistency matters:
• Without consistent methods, bias evaluation becomes ad hoc and unreliable
• Without comparability, we can't determine whether one system is "more biased" than another or whether an intervention improved fairness
• Without repeatability, evaluations can't be verified or contested
• Without transferability, every new system requires inventing a new evaluation approach from scratch

CONTEXT: DIVERSE SYSTEMS AND DOMAINS WITH PATHWAY-DEPENDENT BIAS
The BBS framework provides a potential foundation for consistent evaluation because:
• All mediated systems have pathways, regardless of domain
• All pathways have boundary crossings that can be analyzed for bias entry, transformation, and amplification
• The analytical method (trace the pathway, examine each crossing, assess differential impact) is domain-independent even though the specific findings are domain-specific
• Pathway analysis provides a common vocabulary and analytical structure that different evaluators can apply to different systems

APPLICATION: CENTRAL QUESTION FOR FINAL PROJECT EVALUATION FRAMEWORK
This question connects directly to students' final projects and to the remaining sessions in the course:
• How can the BBS pathway analysis be operationalized into a practical evaluation methodology?
• What criteria should be used at each boundary crossing to assess bias impact?
• How do we weigh different types of bias (entry vs. transformation vs. amplification) in an overall system evaluation?
• How do we handle cases where reducing bias at one boundary crossing increases it at another?

CONNECTION TO COURSE ARC
This slide marks the transition from "understanding bias" to "evaluating bias" — from descriptive analysis to normative methodology. The remaining sessions will build evaluation frameworks that operationalize the BBS-based pathway analysis into practical tools for consistent assessment.

TEACHING NOTE: Frame this as an open question, not a solved problem. Consistent bias evaluation is an active area of research and practice, and there's no consensus methodology. The BBS model provides a promising framework, but translating it into a practical evaluation tool requires exactly the kind of work students will do in their final projects. Let them sit with the difficulty of the question — it's genuine, not rhetorical.

RESPONSIBILITY IS DISTRIBUTED

Purpose: This slide extends the session's central insight about distributed bias to its ethical corollary: if bias is distributed across pathways, then responsibility for that bias must also be distributed across the actors who contribute to those pathways.

NO SINGLE POINT OF ACCOUNTABILITY
On the slide: "Just as bias is distributed across pathways and mediation layers, responsibility cannot be localized to a single actor or decision point in complex AI systems."
Expanded explanation: Our standard accountability frameworks assume that harm can be traced to a responsible party — the person who made the decision, the company that built the product, the developer who wrote the code. But when bias is a pathway property rather than a component property, this attribution becomes deeply problematic:
• The data collector who gathered biased training data may not have known how the data would be used
• The model developer who chose the architecture may not have known what

biases the data contained
• The product manager who set the optimization metric may not have understood its fairness implications
• The deployer who put the system into production may not have had the technical expertise to evaluate its bias properties
• The regulator who approved the system may not have had the tools or authority to trace pathway-level bias

Each actor made locally reasonable decisions. The harm emerged from the combination of those decisions along the pathway. This is the accountability gap that distributed bias creates.

DISTRIBUTED ACROSS STAKEHOLDERS
On the slide: "Accountability is shared among data collectors, system designers, model developers, deployers, auditors, and policymakers — each contributing to outcomes through their role in the system."
Expanded explanation: The BBS pathway model actually helps map responsibility distribution:
• Data collectors are responsible for bias at the data boundary crossing (what gets collected, how, from whom)
• System designers are responsible for bias at the form/interface boundary crossing (what categories exist, what's required, what's excluded)
• Model developers are responsible for bias in the virtual space transformations (what's optimized, how features are engineered, what's amplified)
• Deployers are responsible for bias at the output boundary crossing (where and how the system is applied, what populations are affected)
• Auditors are responsible for identifying bias across the full pathway
• Policymakers are responsible for setting the standards and enforcement mechanisms that govern the entire pathway

This isn't equal distribution — some actors have more influence over critical pathway locations than others. But all contribute, and effective governance requires all to participate.

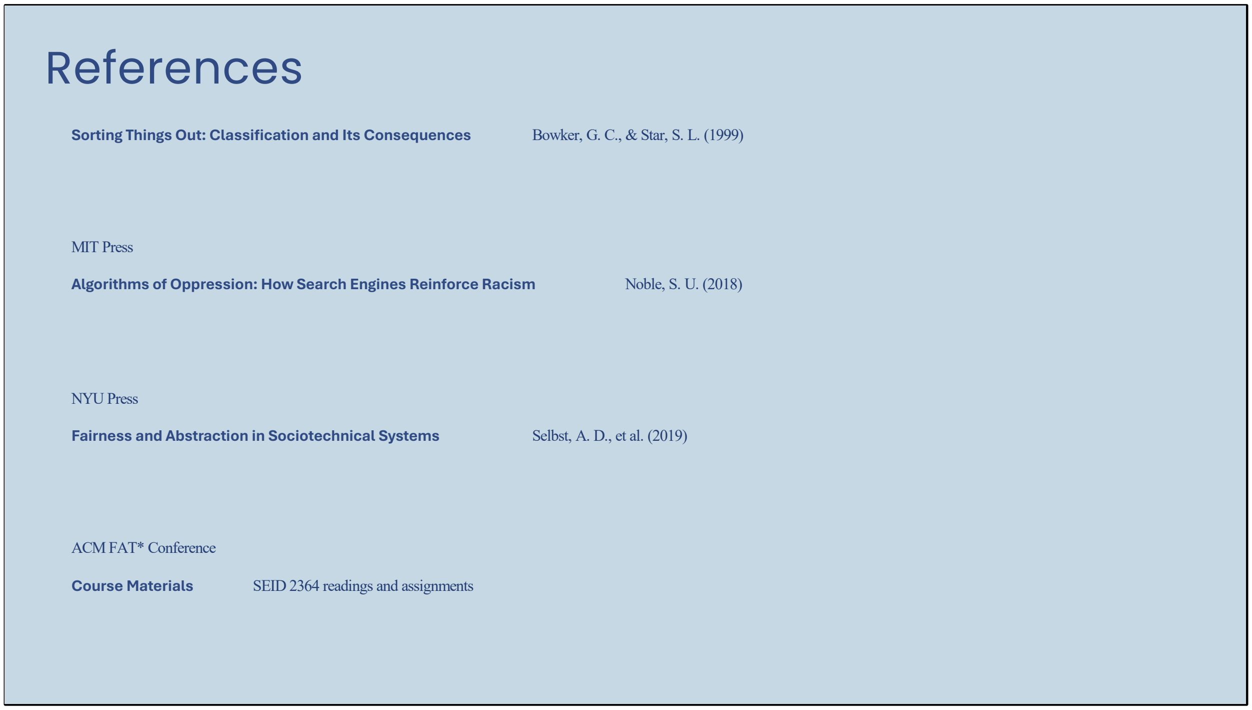
COLLECTIVE RESPONSIBILITY
On the slide: "Addressing bias in AI requires collective action, transparency across system boundaries, and accountability structures that recognize the distributed nature of modern sociotechnical systems."
Expanded explanation: Three requirements for addressing distributed bias:
• Collective action: No single actor can fix pathway-level bias alone. The data collector can't compensate for model architecture choices, and the model developer can't compensate for biased data collection. Effective bias reduction

requires coordinated intervention across the pathway.
• Transparency across boundaries: Each actor needs to understand what happens at other boundary crossings, not just their own. This requires documentation, communication, and shared analytical frameworks (like the BBS model) that allow different actors to understand each other's contributions to the pathway.
• New accountability structures: Legal and organizational frameworks need to evolve beyond "find the responsible party" toward "map the distributed contributions to harm and assign proportional accountability." This is an active area of AI governance research and policy development.

TEACHING NOTE: This slide often generates strong student reactions. Some students find the concept of distributed responsibility philosophically unsatisfying — they want a clear villain. Others worry that distributing responsibility means nobody is accountable. Address both concerns:
• Distributed responsibility doesn't mean nobody's responsible — it means everybody with influence over the pathway bears proportional responsibility
• The BBS model actually makes distributed responsibility more tractable, not less, because it identifies specific boundary crossings where specific actors had specific influence
• The practical challenge isn't philosophical but institutional: building governance structures that can assign and enforce proportional accountability across complex sociotechnical pathways

Assignment Week 8

Students select a new AI system and construct three deployment scenarios to map where bias enters, how it travels through a mediation pathway, and what decisions and harms it produces. Using Stanford Encyclopedia of Philosophy readings on implicit bias, algorithmic fairness, and search ethics as analytical tools, students trace bias locations at every stage of the pathway — from data source to consequential decision — and compare how deployment context changes the risk.



REFERENCES AND FURTHER READING

These references provide the theoretical foundations for this session's content:

BOWKER, G. C., & STAR, S. L. (1999). SORTING THINGS OUT: CLASSIFICATION AND ITS CONSEQUENCES. MIT PRESS.
This is the foundational text on how classification systems shape social reality. Bowker and Star's analysis of how categories are created, maintained, and enforced is directly relevant to this session's discussion of how form design, encoding choices, and categorization decisions at boundary objects introduce bias. Their concept of "residual categories" (the catch-all "Other" option) is particularly relevant to the discussion of dropdown menus and categorization on slide 24. Key insight: classification is never a neutral technical act — it is always a political and ethical one that advantages some and disadvantages others.

NOBLE, S. U. (2018). ALGORITHMS OF OPPRESSION: HOW SEARCH ENGINES REINFORCE RACISM. NYU PRESS.
Noble's work demonstrates how commercial search algorithms reproduce and amplify racial stereotypes and discrimination. This is a concrete case study of

the contemporary AI bias described on slide 16 — bias that is distributed, emergent, and operates through multiple mediation layers (query interpretation, index construction, ranking algorithms, advertising placement). Noble's analysis illustrates pathway-level bias that no single component audit would detect. Key insight: algorithmic bias is not a technical malfunction but a structural property of systems designed within and for a society organized around racial hierarchy.

SELBST, A. D., ET AL. (2019). FAIRNESS AND ABSTRACTION IN SOCIOTECHNICAL SYSTEMS. ACM FAT* CONFERENCE.
This paper directly addresses the challenge of consistent evaluation raised on slide 32. Selbst et al. identify five "traps" that technical approaches to algorithmic fairness fall into, all of which involve abstracting away the sociotechnical context in which systems operate. Their analysis supports this session's argument that bias must be analyzed at the pathway level (not the component level) and that evaluation must account for the full sociotechnical context (not just the algorithm). Key insight: fairness cannot be reduced to a mathematical property of an algorithm — it is a property of the entire sociotechnical system, including its deployment context, affected populations, and governance structures.

COURSE MATERIALS: SEID 2364 READINGS AND ASSIGNMENTS
Students should revisit their Session 7 readings on ethical reasoning frameworks in light of this session's structural analysis. The move from individual ethical reasoning to systemic bias analysis doesn't replace earlier frameworks — it adds a structural dimension that ethical reasoning alone cannot provide. The BBS model and pathway analysis are tools that extend ethical reasoning into complex sociotechnical systems where individual moral judgment is necessary but insufficient.

SUPPLEMENTARY READING SUGGESTIONS:
• Eubanks, V. (2018). Automating Inequality. St. Martin's Press. — Detailed case studies of how automated systems in public services reproduce and deepen poverty and inequality.
• Benjamin, R. (2019). Race After Technology. Polity Press. — Analysis of how emerging technologies encode existing racial hierarchies while appearing neutral.
• Barocas, S., & Selbst, A. D. (2016). Big Data's Disparate Impact. California Law Review. — Legal analysis of how data mining can produce discrimination, even without discriminatory intent.

TEACHING NOTE: Encourage students to engage with at least one of these references in depth for their final projects. The theoretical frameworks they provide are essential for moving beyond "I found bias in my system" to "I can analyze, explain, and propose interventions for the structural mechanisms that produce bias in my system."